

where $K_c = 128$. This is because

$$K_v = \lim_{s \rightarrow 0} s G_c(s) G(s) = \lim_{s \rightarrow 0} s K_c G(s) = K_c \frac{10}{16} = 80$$

and we obtain $K_c = 128$. The angle deficiency at the desired closed-loop pole $s = -2 + j2\sqrt{3}$ is

$$\text{Angle deficiency} = 120^\circ + 90^\circ + 30^\circ - 180^\circ = 60^\circ$$

The lead portion of the lag-lead compensator must contribute this angle. To choose T_1 we may use the graphical method presented in Section 7-5.

The lead portion must satisfy the following conditions:

$$\left| 128 \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\beta}{T_1}} G(s_1) \right|_{s_1 = -2 + j2\sqrt{3}} = 1$$

and

$$\angle \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\beta}{T_1}} \bigg|_{s_1 = -2 + j2\sqrt{3}} = 60^\circ$$

The first condition can be simplified as

$$\left| \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\beta}{T_1}} \right|_{s_1 = -2 + j2\sqrt{3}} = \frac{1}{13.3333}$$

By using the same approach as used in Section 7-5, the zero ($s = 1/T_1$) and pole ($s = \beta/T_1$) can be determined as follows:

$$\frac{1}{T_1} = 3.70, \quad \frac{\beta}{T_1} = 53.35$$

See Figure 7-34. The value of β is thus determined as

$$\beta = 14.419$$

For the lag portion of the compensator, we may choose

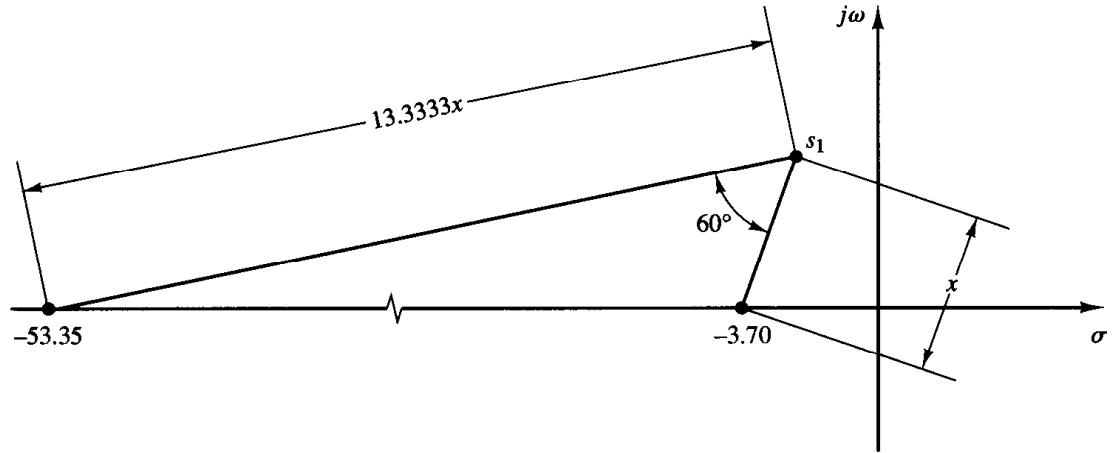
$$\frac{1}{\beta T_2} = 0.01$$

Then

$$\frac{1}{T_2} = 0.1442$$

Noting that

Figure 7–34
Graphical determination of the zero and pole of the lead portion of the compensator.



$$\left| \frac{s_1 + 0.1442}{s_1 + 0.01} \right|_{s_1 = -2 + j2\sqrt{3}} = 0.9837$$

$$\angle \frac{s_1 + 0.1442}{s_1 + 0.01} \Big|_{s_1 = -2 + j2\sqrt{3}} = -1.697^\circ$$

the angle contribution of the lag portion is -1.697° and the magnitude contribution is 0.9837. This means that the dominant closed-loop poles lie close to the desired location $s = -2 \pm j2\sqrt{3}$. Thus the compensator designed,

$$G_c(s) = 128 \left(\frac{s + 3.70}{s + 53.35} \right) \left(\frac{s + 0.1442}{s + 0.01} \right)$$

is acceptable. The feedforward transfer function of the compensated system becomes

$$G_c(s)G(s) = \frac{1280(s + 3.7)(s + 0.1442)}{s(s + 53.35)(s + 0.01)(s + 2)(s + 8)}$$

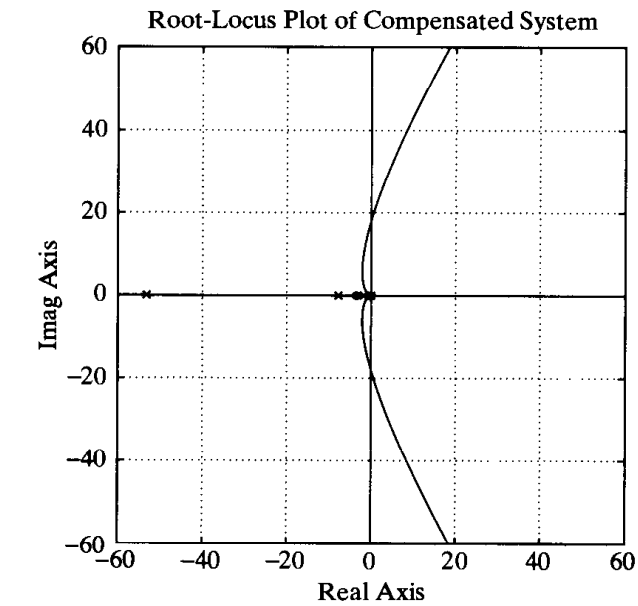
A root-locus plot of the compensated system is shown in Figure 7–35(a). An enlarged root-locus plot near the origin is shown in Figure 7–35(b).

To verify the improved system performance of the compensated system, see the unit-step responses and unit-ramp responses of the compensated and uncompensated systems shown in Figures 7–36 (a) and (b), respectively.

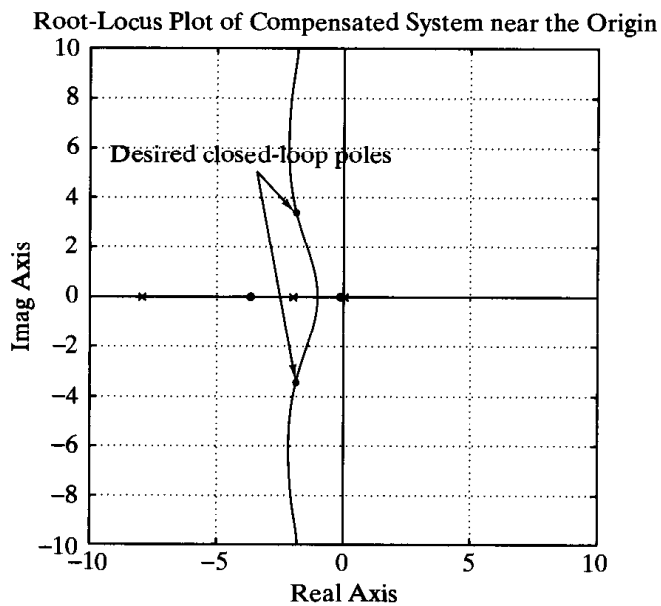
- A-7-9.** Consider the system shown in Figure 7–37. Design a lag–lead compensator such that the static velocity error constant K_v is 50 sec^{-1} and the damping ratio ζ of the dominant closed-loop poles is 0.5. (Choose the zero of the lead portion of the lag–lead compensator to cancel the pole at $s = -1$ of the plant.) Determine all closed-loop poles of the compensated system.

Solution. Let us employ the lag–lead compensator given by

$$G_c(s) = K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\beta}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) = K_c \frac{(T_1 s + 1)(T_2 s + 1)}{\left(\frac{T_1}{\beta} s + 1 \right) (\beta T_2 s + 1)}$$



(a)



(b)

Figure 7-35

(a) Root-locus plot of compensated system; (b) root-locus plot near the origin.

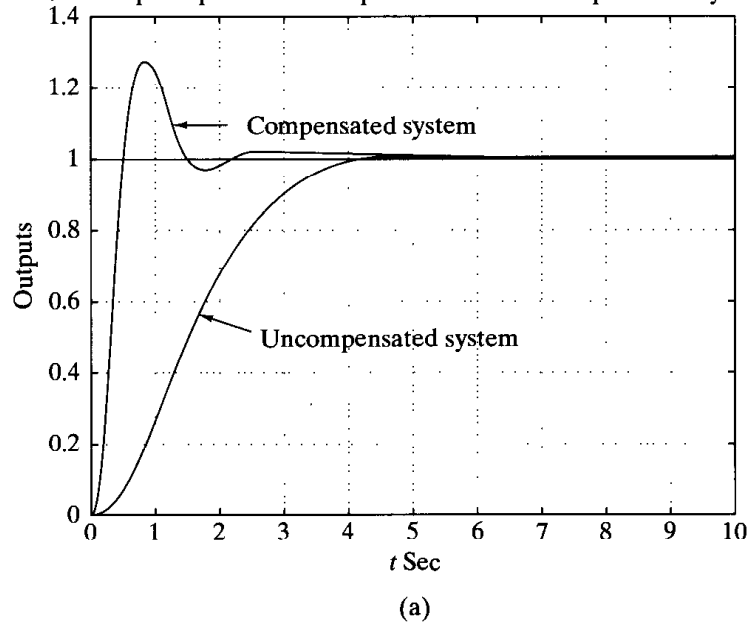
where $\beta > 1$. Then

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} s G_c(s) G(s) \\ &= \lim_{s \rightarrow 0} s \frac{K_c (T_1 s + 1)(T_2 s + 1)}{\left(\frac{T_1}{\beta} s + 1\right) (\beta T_2 s + 1)} \frac{1}{s(s+1)(s+5)} \\ &= \frac{K_c}{5} \end{aligned}$$

The specification that $K_v = 50 \text{ sec}^{-1}$ determines the value of K_c , or

$$K_c = 250$$

Unit-Step Responses of Compensated and Uncompensated Systems



Unit-Ramp Responses of Compensated and Uncompensated Systems

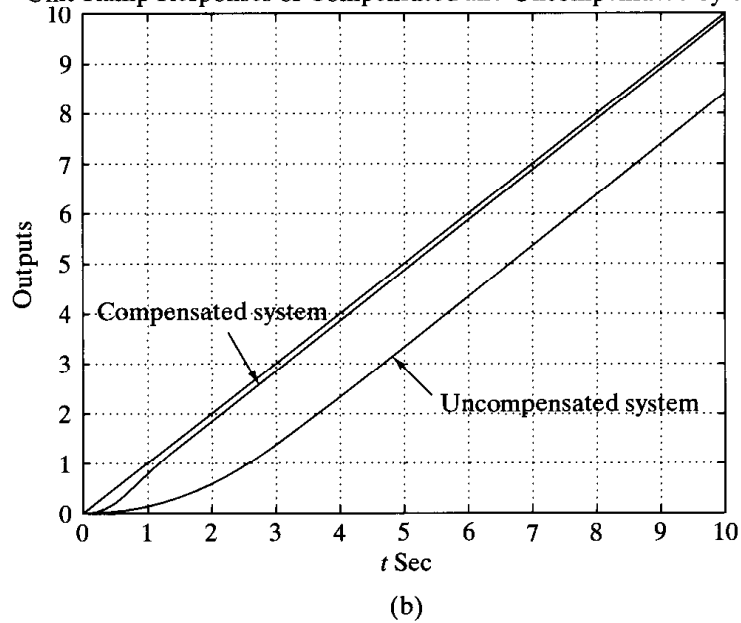


Figure 7-36
(a) Unit-step responses of compensated and uncompensated systems;
(b) unit-ramp responses of both systems.

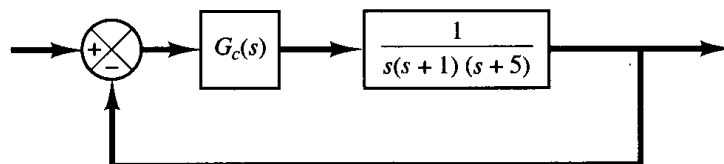


Figure 7-37
Control system.

We now choose $T_1 = 1$ so that $s + (1/T_1)$ will cancel the $(s + 1)$ term of the plant. The lead portion then becomes

$$\frac{s + 1}{s + \beta}$$

For the lag portion of the lag-lead compensator we require

$$\left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right| \doteq 1, \quad -5^\circ < \angle \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} < 0^\circ$$

where $s = s_1$ is one of the dominant closed-loop poles. For $s = s_1$, the open-loop transfer function becomes

$$G_c(s_1)G(s_1) \doteq K_c \left(\frac{s_1 + 1}{s_1 + \beta} \right) \frac{1}{s_1(s_1 + 1)(s_1 + 5)} = K_c \frac{1}{(s_1 + \beta)s_1(s_1 + 5)}$$

Noting that at $s = s_1$ the magnitude and angle conditions are satisfied, we have

$$\left| K_c \frac{1}{s_1(s_1 + \beta)(s_1 + 5)} \right| = 1 \quad (7-8)$$

$$\angle K_c \frac{1}{s_1(s + \beta)(s_1 + 5)} = \pm 180^\circ(2k + 1) \quad (7-9)$$

where $k = 0, 1, 2, \dots$. In Equations (7-8) and (7-9), β and s_1 are unknowns. Since the damping ratio ζ of the dominant closed-loop poles is specified as 0.5, the closed-loop pole $s = s_1$ can be written as

$$s_1 = -x + j\sqrt{3}x$$

where x is as yet undetermined.

Notice that the magnitude condition, Equation (7-8), can be rewritten as

$$\left| \frac{K_c}{(-x + j\sqrt{3}x)(-x + \beta + j\sqrt{3}x)(-x + 5 + j\sqrt{3}x)} \right| = 1$$

Noting that $K_c = 250$, we have

$$x\sqrt{(\beta - x)^2 + 3x^2} \sqrt{(5 - x)^2 + 3x^2} = 125 \quad (7-10)$$

The angle condition, Equation (7-9), can be rewritten as

$$\begin{aligned} & \angle K_c \frac{1}{(-x + j\sqrt{3}x)(-x + \beta + j\sqrt{3}x)(-x + 5 + j\sqrt{3}x)} \\ &= -120^\circ - \tan^{-1} \left(\frac{\sqrt{3}x}{-x + \beta} \right) - \tan^{-1} \left(\frac{\sqrt{3}x}{-x + 5} \right) = -180^\circ \end{aligned}$$

or

$$\tan^{-1} \left(\frac{\sqrt{3}x}{-x + \beta} \right) + \tan^{-1} \left(\frac{\sqrt{3}x}{-x + 5} \right) = 60^\circ \quad (7-11)$$

We need to solve Equations (7-10) and (7-11) for β and x . By several trial-and-error calculations, it can be found that

$$\beta = 16.025, \quad x = 1.9054$$

Thus

$$s_1 = -1.9054 + j\sqrt{3}(1.9054) = -1.9054 + j3.3002$$

The lag portion of the lag-lead compensator can be determined as follows: Noting that the pole and zero of the lag portion of the compensator must be located near the origin, we may choose

$$\frac{1}{\beta T_2} = 0.01$$

That is,

$$\frac{1}{T_2} = 0.16025 \quad \text{or} \quad T_2 = 6.25$$

With the choice of $T_2 = 6.25$, we find

$$\begin{aligned} \left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right| &= \left| \frac{-1.9054 + j3.3002 + 0.16025}{-1.9054 + j3.3002 + 0.01} \right| \\ &= \left| \frac{-1.74515 + j3.3002}{-1.89054 + j3.3002} \right| = 0.98 \div 1 \end{aligned} \quad (7-12)$$

and

$$\begin{aligned} \angle \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} &= \angle \frac{-1.9054 + j3.3002 + 0.16025}{-1.9054 + j3.3002 + 0.01} \\ &= \tan^{-1} \left(\frac{3.3002}{-1.74515} \right) - \tan^{-1} \left(\frac{3.3002}{-1.89054} \right) = -1.937^\circ \end{aligned} \quad (7-13)$$

Since

$$-5^\circ < -1.937^\circ < 0^\circ$$

our choice of $T_2 = 6.25$ is acceptable. Then the lag-lead compensator just designed can be written as

$$G_c(s) = 250 \left(\frac{s + 1}{s + 16.025} \right) \left(\frac{s + 0.16025}{s + 0.01} \right)$$

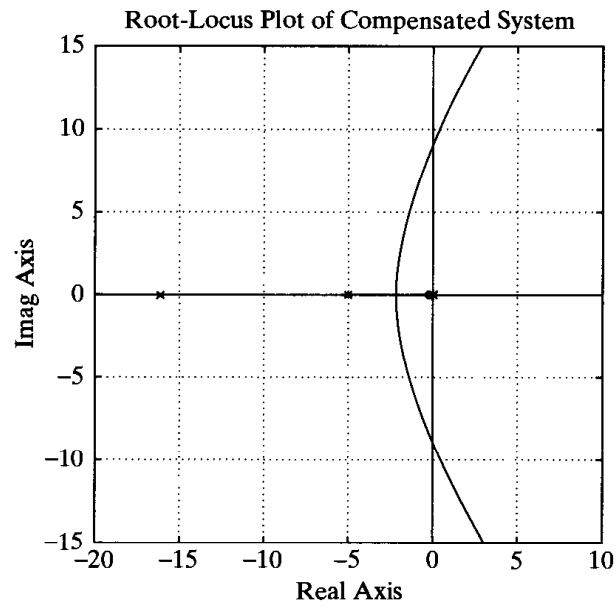
Therefore, the compensated system has the following open-loop transfer function:

$$G_c(s)G(s) = \frac{250(s + 0.16025)}{s(s + 0.01)(s + 5)(s + 16.025)}$$

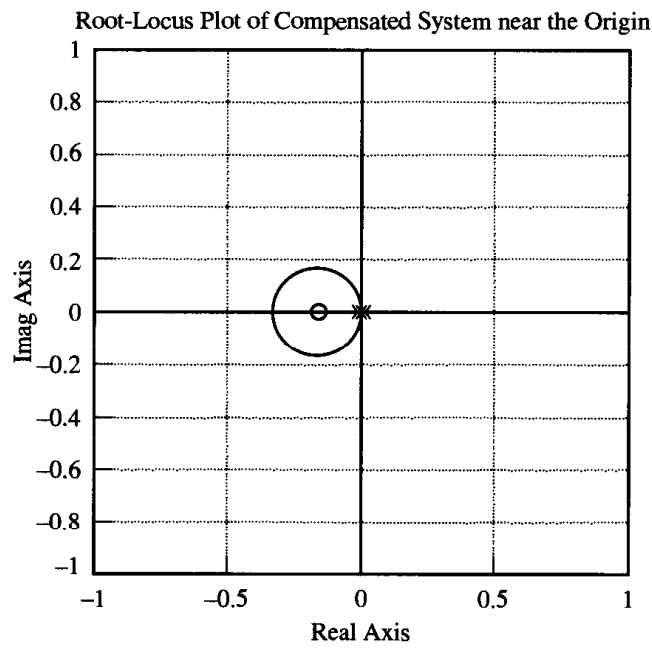
A root-locus plot of the compensated system is shown in Figure 7-38(a). An enlarged root-locus plot near the origin is shown in Figure 7-38(b).

The closed loop transfer function becomes

$$\frac{C(s)}{R(s)} = \frac{250(s + 0.16025)}{s(s + 0.01)(s + 5)(s + 16.025) + 250(s + 0.16025)}$$



(a)



(b)

Figure 7-38

(a) Root-locus plot of compensated system; (b) root-locus plot near the origin.

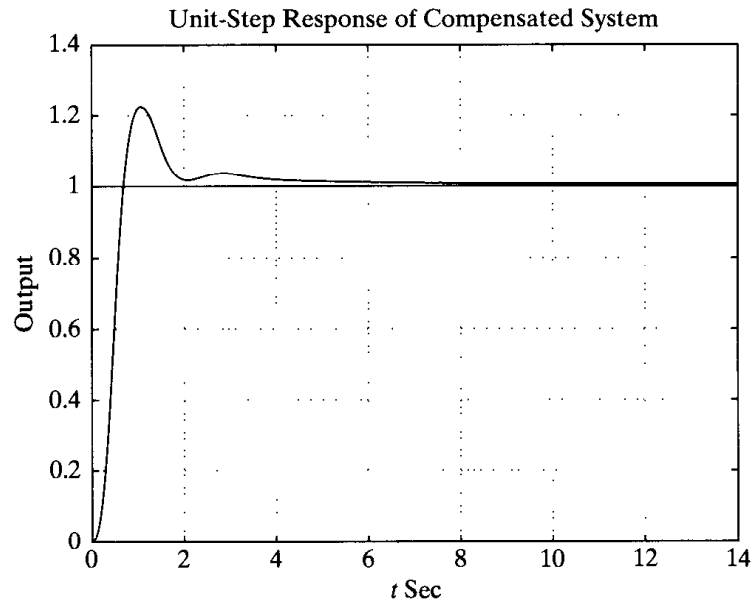
The closed-loop poles are located at

$$s = -1.8308 \pm j3.2359$$

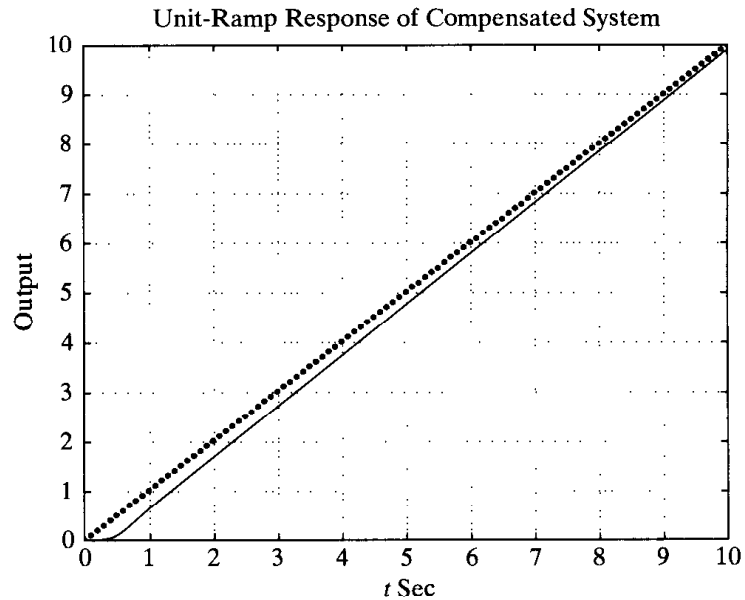
$$s = -0.1684$$

$$s = -17.205$$

Notice that the dominant closed-loop poles $s = -1.8308 \pm j3.2359$ differ from the dominant closed-loop poles $s = \pm s_1$ assumed in the computation of β and T_2 . Small deviations of the dominant closed-loop poles $s = -1.8308 \pm j3.2359$ from $s = \pm s_1 = -1.9054 \pm j3.3002$ are due to the approximations involved in determining the lag portion of the compensator [See Equations (7-12) and (7-13)].



(a)



(b)

Figure 7-39

(a) Unit-step response of the compensated system;
(b) unit-ramp response of the compensated system.

Figures 7-39(a) and (b) show the unit-step response and unit-ramp response of the designed system, respectively. Note that the closed-loop pole at $s = -0.1684$ almost cancels the zero at $s = -0.16025$. However, this pair of closed-loop pole and zero located near the origin produces a long tail of small amplitude. Since the closed-loop pole at $s = -17.205$ is located very much farther to the left compared to the closed-loop poles at $s = -1.8308 \pm j3.2359$, the effect of this real pole on the system response is also very small. Therefore, the closed-loop poles at $s = -1.8308 \pm j3.2359$ are indeed dominant closed-loop poles that determine the response characteristics of the closed-loop system. In the unit-ramp response, the steady-state error in following the unit-ramp input eventually becomes $1/K_v = \frac{1}{50} = 0.02$.

A-7-10. Consider the system shown in Figure 7-40. It is desired to design a PID controller $G_c(s)$ such that the dominant closed-loop poles are located at $s = -1 \pm j\sqrt{3}$. For the PID controller,

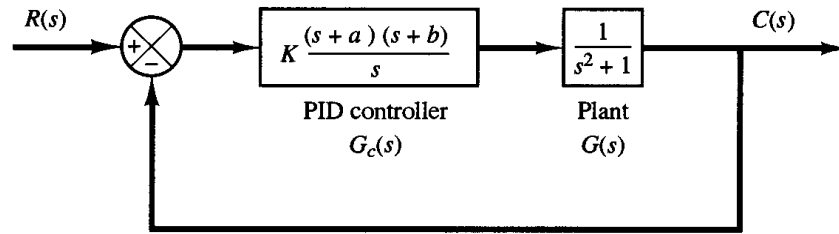


Figure 7–40
PID-controlled
system.

choose $a = 1$ and then determine the values of K and b . Sketch the root-locus diagram for the designed system.

Solution. Since

$$G_c(s)G(s) = K \frac{(s+1)(s+b)}{s} \frac{1}{s^2+1}$$

the sum of the angles at $s = -1 + j\sqrt{3}$, one of the desired closed-loop poles, from the zero at $s = -1$ and poles at $s = 0, s = j$, and $s = -j$ is

$$90^\circ - 143.794^\circ - 120^\circ - 110.104^\circ = -283.898^\circ$$

Hence the zero at $s = -b$ must contribute 103.898° . This requires that the zero be located at

$$b = 0.5714$$

The gain constant K can be determined from the magnitude condition.

$$\left| K \frac{(s+1)(s+0.5714)}{s} \frac{1}{s^2+1} \right|_{s=-1+j\sqrt{3}} = 1$$

or

$$K = 2.3333$$

Then the compensator can be written as follows:

$$G_c(s) = 2.3333 \frac{(s+1)(s+0.5714)}{s}$$

The open-loop transfer function becomes

$$G_c(s)G(s) = \frac{2.3333(s+1)(s+0.5714)}{s} \frac{1}{s^2+1}$$

From this equation a root-locus plot for the compensated system can be drawn. Figure 7–41 is a root-locus plot.

The closed-loop transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{2.3333(s+1)(s+0.5714)}{s^3 + s + 2.3333(s+1)(s+0.5714)}$$

The closed-loop poles are located at $s = -1 \pm j\sqrt{3}$ and $s = -0.3333$. A unit-step response curve is shown in Figure 7–42. The closed-loop pole at $s = -0.3333$ and a zero at $s = -0.5714$ produces a long tail of small amplitude.

- A-7-11.** Figure 7–43(a) is a block diagram of a model for an attitude-rate control system. The closed-loop transfer function for this system is

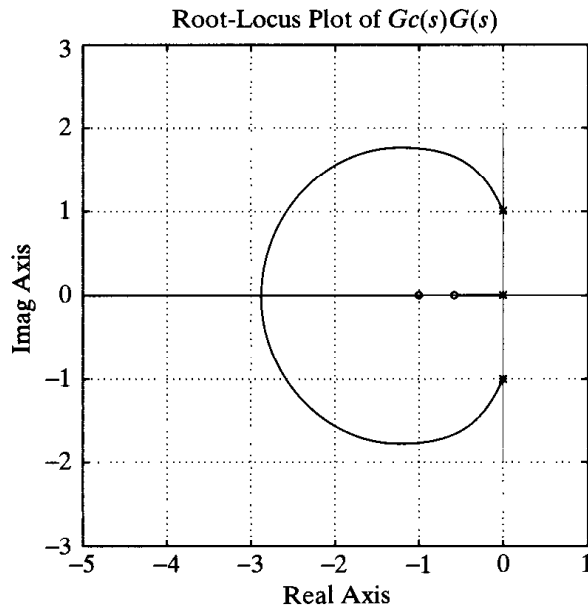


Figure 7-41
Root-locus plot
of the compen-
sated system (Prob-
lem A-7-10).

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{2s + 0.1}{s^3 + 0.1s^2 + 6s + 0.1} \\ &= \frac{2(s + 0.05)}{(s + 0.0417 + j2.4489)(s + 0.0417 - j2.4489)(s + 0.0167)}\end{aligned}$$

The unit-step response of this system is shown in Figure 7-43(b). The response shows high-frequency oscillations at the beginning of the response due to the poles at $s = -0.0417 \pm j2.4489$. The response is dominated by the pole at $s = -0.0167$. The settling time is approximately 240 sec.

It is desired to speed up the response and also eliminate the oscillatory mode at the beginning of the response. Design a suitable compensator such that the dominant closed-loop poles are at $s = -2 \pm j2\sqrt{3}$.

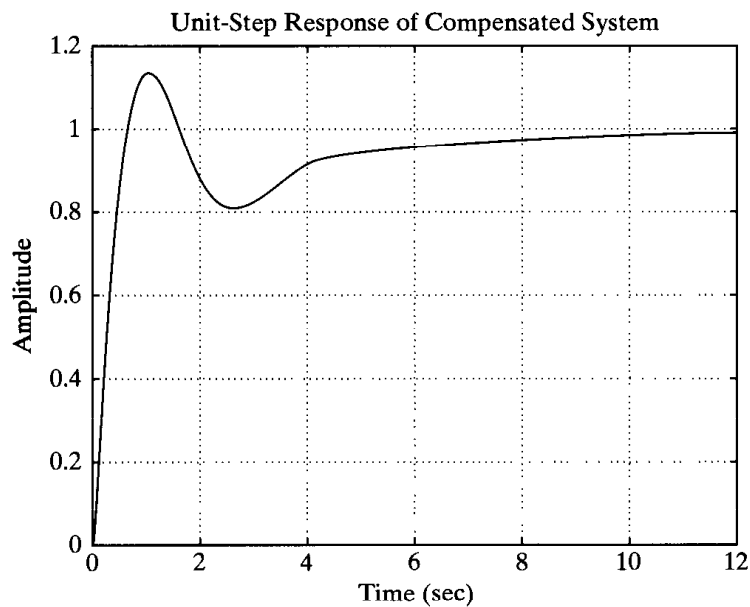


Figure 7-42
Unit-step response
of the compensated
system (Problem
A-7-10).

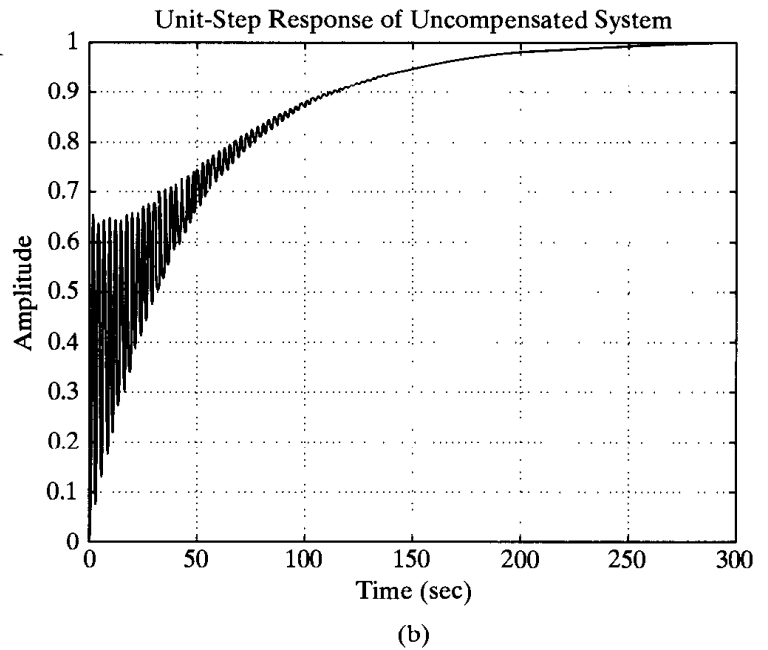
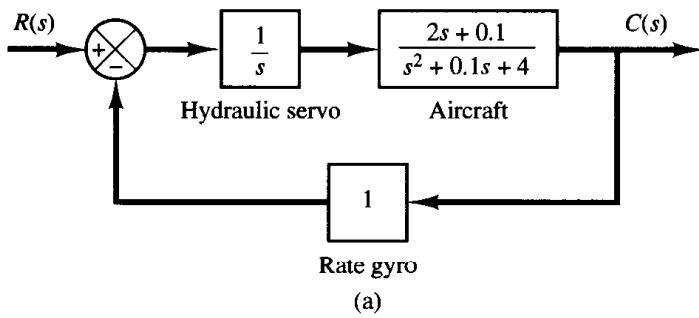


Figure 7-43
(a) Attitude-rate control system;
(b) unit-step response.

Solution. Figure 7-44 shows a block diagram for the compensated system. Note that the open-loop zero at $s = -0.05$ and the open-loop pole at $s = 0$ generate a closed-loop pole between $s = 0$ and $s = -0.05$. Such a closed-loop pole becomes a dominant closed-loop pole and make the response quite slow. Hence, it is necessary to replace this zero by a zero that is located far away from the $j\omega$ axis, for example, a zero at $s = -4$.

We now choose the compensator in the following form:

$$G_c(s) = \hat{G}_c(s) \frac{s + 4}{2s + 0.1}$$

Then the open-loop transfer function of the compensated system becomes

$$\begin{aligned} G_c(s)G(s) &= \hat{G}_c(s) \frac{s + 4}{2s + 0.1} \frac{1}{s} \frac{2s + 0.1}{s^2 + 0.1s + 4} \\ &= \hat{G}_c(s) \frac{s + 4}{s(s^2 + 0.1s + 4)} \end{aligned}$$

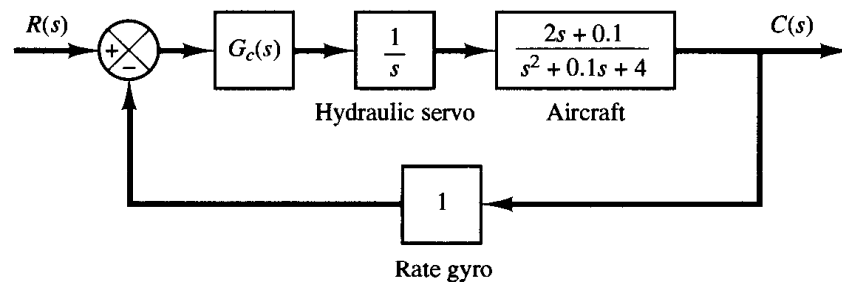


Figure 7-44
Compensated attitude-rate control system.

To determine $\hat{G}_c(s)$ by the root-locus method, we need to find the angle deficiency at the desired closed-loop pole $s = -2 + j2\sqrt{3}$. The angle deficiency can be found as follows:

$$\begin{aligned}\text{Angle deficiency} &= -143.088^\circ - 120^\circ - 109.642^\circ + 60^\circ + 180^\circ \\ &= -132.73^\circ\end{aligned}$$

Hence, the lead compensator $\hat{G}_c(s)$ must provide 132.73° . Since the angle deficiency is 132.73° , we need two lead compensators, each providing 66.365° . Thus $G_c(s)$ will have the following form:

$$G_c(s) = K_c \left(\frac{s + s_z}{s + s_p} \right)^2$$

Suppose that we choose two zeros at $s = -2$. Then the two poles of the lead compensators can be obtained from

$$\frac{3.4641}{s_p - 2} = \tan(90^\circ - 66.365^\circ) = 0.4376169$$

or

$$\begin{aligned}s_p &= 2 + \frac{3.4641}{0.4376169} \\ &= 9.9158\end{aligned}$$

Hence,

$$\hat{G}_c(s) = K_c \left(\frac{s + 2}{s + 9.9158} \right)^2$$

The entire compensator $G_c(s)$ for the system becomes

$$G_c(s) = \hat{G}_c(s) \frac{s + 4}{2s + 0.1} = K_c \frac{(s + 2)^2}{(s + 9.9158)^2} \frac{s + 4}{2s + 0.1}$$

The value of K_c can be determined from the magnitude condition. Since the open-loop transfer function is

$$G_c(s)G(s) = K_c \frac{(s + 2)^2(s + 4)}{(s + 9.9158)^2 s(s^2 + 0.1s + 4)}$$

the magnitude condition becomes

$$\left| K_c \frac{(s + 2)^2(s + 4)}{(s + 9.9158)^2 s(s^2 + 0.1s + 4)} \right|_{s=-2+j2\sqrt{3}} = 1$$

Hence,

$$\begin{aligned}K_c &= \left| \frac{(s + 9.9158)^2 s(s^2 + 0.1s + 4)}{(s + 2)^2(s + 4)} \right|_{s=-2+j2\sqrt{3}} \\ &= 88.0227\end{aligned}$$

Thus the compensator $G_c(s)$ becomes

$$G_c(s) = 88.0227 \frac{(s + 2)^2(s + 4)}{(s + 9.9158)^2(2s + 0.1)}$$

The open-loop transfer function is given by

$$G_c(s)G(s) = \frac{88.0227(s + 2)^2(s + 4)}{(s + 9.9158)^2s(s^2 + 0.1s + 4)}$$

A root-locus plot for the compensated system is shown in Figure 7–45. The closed-loop poles for the compensated system are indicated in the plot. The closed-loop poles, the roots of the characteristic equation

$$(s + 9.9158)^2s(s^2 + 0.1s + 4) + 88.0227(s + 2)^2(s + 4) = 0$$

are as follows:

$$s = -2.0000 \pm j3.4641$$

$$s = -7.5224 \pm j6.5326$$

$$s = -0.8868$$

Now that the compensator has been designed, we shall examine the transient response characteristics with MATLAB. The closed-loop transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{88.0227(s + 2)^2(s + 4)}{(s + 9.9158)^2s(s^2 + 0.1s + 4) + 88.0227(s + 2)^2(s + 4)}$$

Figures 7–46(a) and (b) show the plots of the unit-step response and unit-ramp response of the compensated system. These response curves show that the designed system is acceptable.

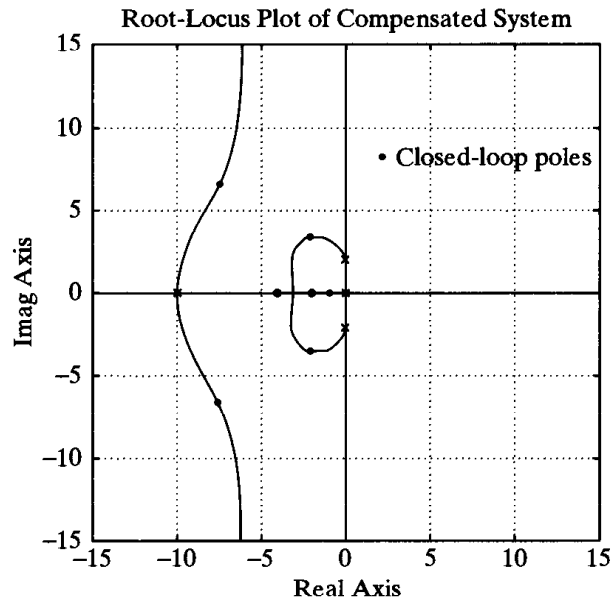


Figure 7–45
Root-locus plot
of the compen-
sated system.

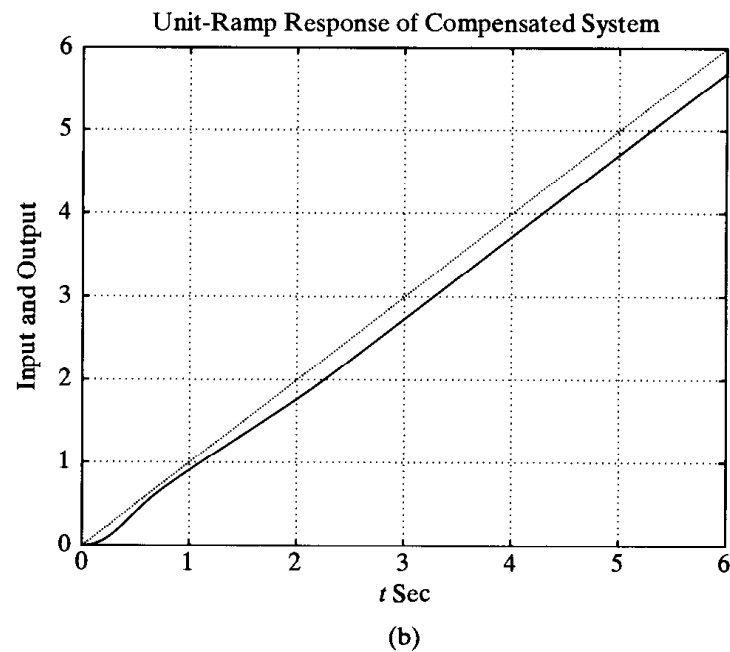
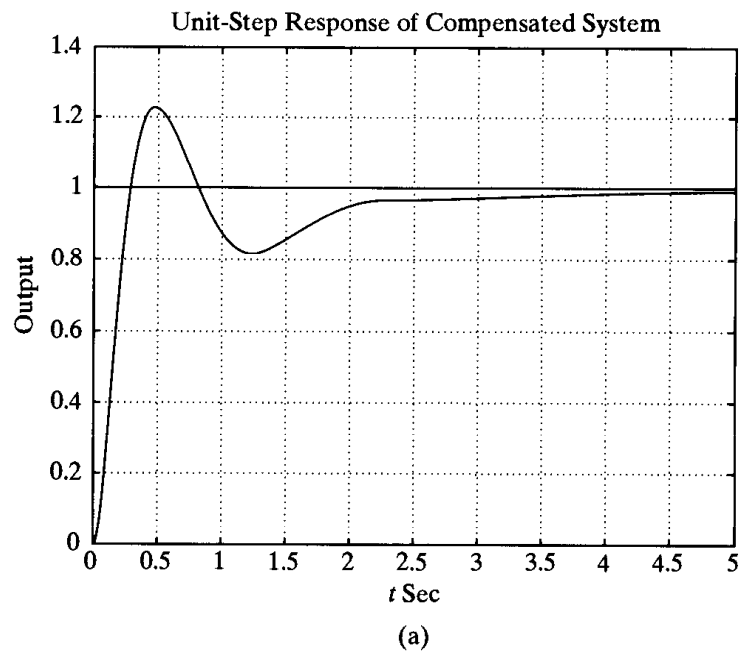


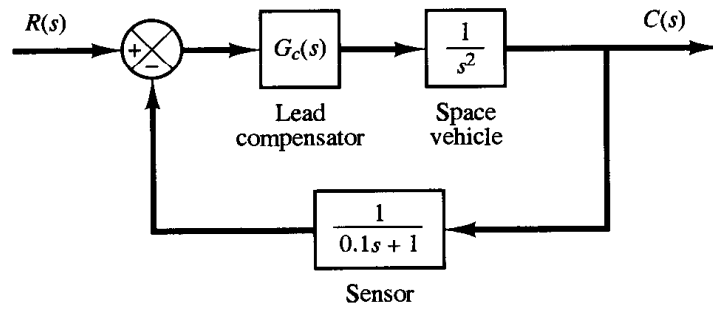
Figure 7-46
 (a) Unit-step re-
 sponse of the com-
 pensated system;
 (b) unit-ramp re-
 sponse of the com-
 pensated system
 (Problem A-7-11).

- A-7-12.** Consider the model for a space vehicle control system shown in Figure 7-47. Design a lead compensator $G_c(s)$ such that the damping ratio ζ and the undamped natural frequency ω_n of the dominant closed-loop poles are 0.5 and 2 rad/sec, respectively.

Solution

First attempt: Assume the lead compensator $G_c(s)$ to be

Figure 7-47
Space vehicle control system.



$$G_c(s) = K_c \left(\frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} \right) \quad (0 < \alpha < 1)$$

From the given specifications, $\zeta = 0.5$ and $\omega_n = 2$ rad/sec, the dominant closed-loop poles must be located at

$$s = -1 \pm j\sqrt{3}$$

We first calculate the angle deficiency at this closed-loop pole.

$$\begin{aligned} \text{Angle deficiency} &= -120^\circ - 120^\circ - 10.8934^\circ + 180^\circ \\ &= -70.8934^\circ \end{aligned}$$

This angle deficiency must be compensated by the lead compensator. There are many ways to determine the locations of the pole and zero of the lead network. Let us choose the zero of the compensator at $s = -1$. Then, referring to Figure 7-48, we have the following equation:

$$\frac{1.73205}{x - 1} = \tan(90^\circ - 70.8934^\circ) = 0.34641$$

or

$$x = 1 + \frac{1.73205}{0.34641} = 6$$

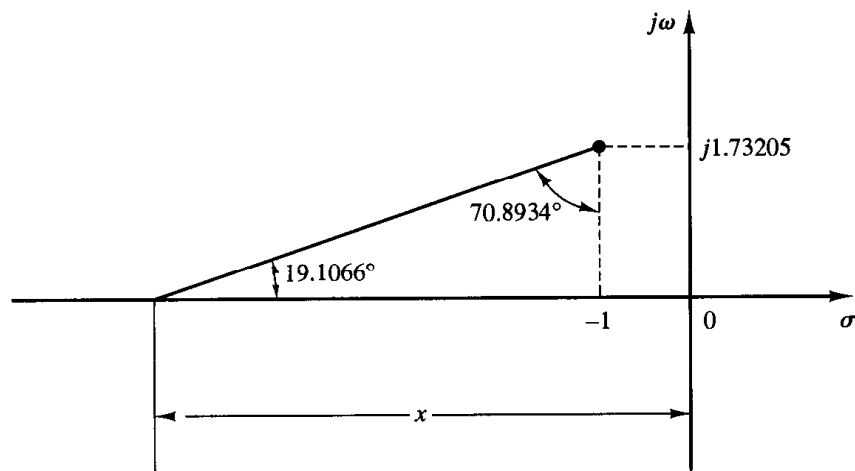


Figure 7-48
Determination of the pole of the lead network.

Hence,

$$G_c(s) = K_c \frac{s+1}{s+6}$$

The value of K_c can be determined from the magnitude condition

$$K_c \left| \frac{s+1}{s+6} \frac{1}{s^2} \frac{1}{0.1s+1} \right|_{s=-1+j\sqrt{3}} = 1$$

as follows:

$$K_c = \left| \frac{(s+6)s^2(0.1s+1)}{s+1} \right|_{s=-1+j\sqrt{3}} = 11.2000$$

Thus

$$G_c(s) = 11.2 \frac{s+1}{s+6}$$

Since the open-loop transfer function becomes

$$\begin{aligned} G_c(s)G(s)H(s) &= 11.2 \frac{s+1}{(s+6)s^2(0.1s+1)} \\ &= \frac{11.2(s+1)}{0.1s^4 + 1.6s^3 + 6s^2} \end{aligned}$$

a root-locus plot of the compensated system can be obtained easily with MATLAB by entering num and den and using rlocus command. The result is shown in Figure 7-49.

The closed-loop transfer function for the compensated system becomes

$$\frac{C(s)}{R(s)} = \frac{11.2(s+1)(0.1s+1)}{(s+6)s^2(0.1s+1) + 11.2(s+1)}$$

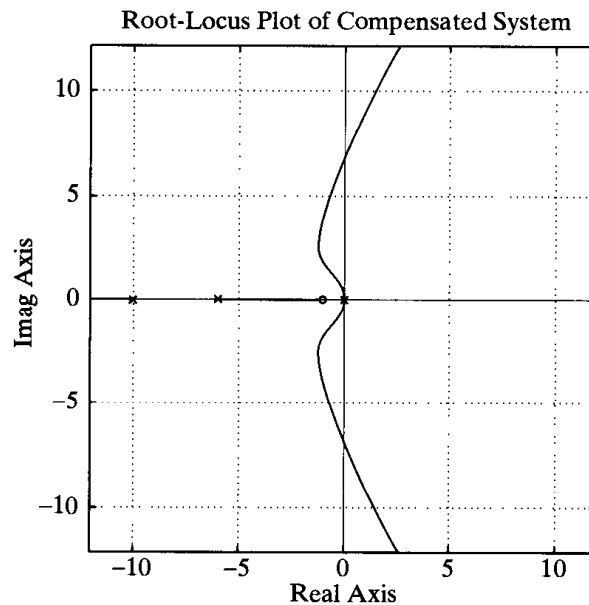


Figure 7-49
Root-locus plot
of the compen-
sated system.

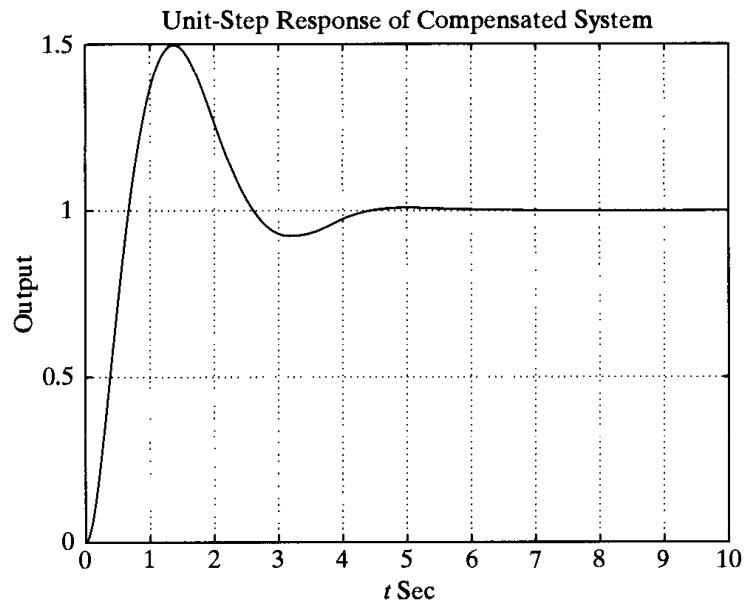


Figure 7-50
Unit-step response
of the compensated
system.

Figure 7-50 shows the unit-step response curve. It shows a fairly large overshoot (50% overshoot). It is desirable to modify the compensator and make the maximum overshoot smaller. A close look at the root-locus plot reveals that the presence of the zero at $s = -1$ is adding the amount of the maximum overshoot. One way to avoid this situation is to modify the lead compensator as presented in the following second attempt.

Second attempt: To modify the shape of the root loci, we may use two lead networks, each contributing half the necessary lead angle, which is $70.8934^\circ/2 = 35.4467^\circ$. Let us choose the location of the zeros at $s = -3$. (This is an arbitrary choice. Other choices such as $s = -2.5$ and $s = -4$ may be made.)

Once we choose two zeros at $s = -3$, the necessary location of the poles can be determined as shown in Figure 7-51, or

$$\begin{aligned} \frac{1.73205}{y - 1} &= \tan(40.89334^\circ - 35.4467^\circ) \\ &= \tan 5.4466^\circ = 0.09535 \end{aligned}$$

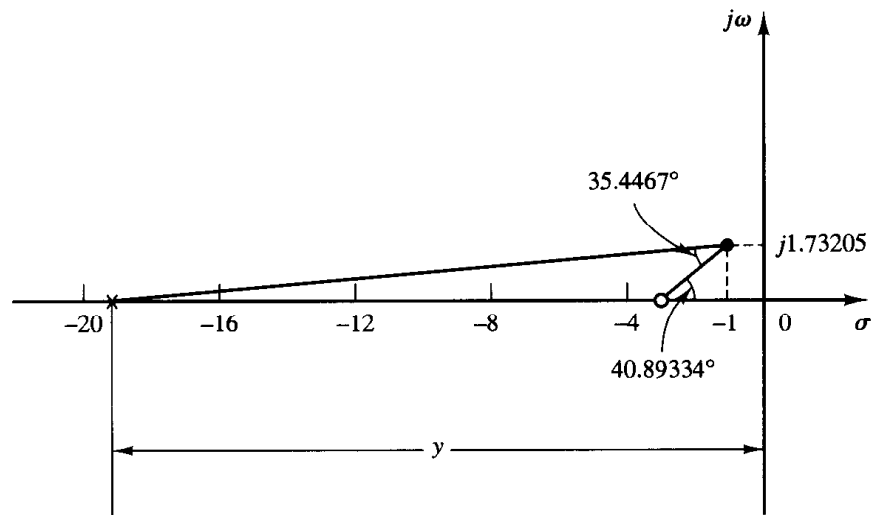


Figure 7-51
Determination of
the pole of the
lead network.

which yields

$$y = 1 + \frac{1.73205}{0.09535} = 19.1652$$

Hence, the lead compensator will have the following transfer function:

$$G_c(s) = K_c \left(\frac{s + 3}{s + 19.1652} \right)^2$$

The value of K_c can be determined from the magnitude condition as follows:

$$\left| K_c \left(\frac{s + 3}{s + 19.1652} \right)^2 \frac{1}{s^2} \frac{1}{0.1s + 1} \right|_{s=-1+j\sqrt{3}} = 1$$

or

$$K_c = 174.3864$$

Then the lead compensator just designed is

$$G_c(s) = 174.3864 \left(\frac{s + 3}{s + 19.1652} \right)^2$$

Then the open-loop transfer function becomes

$$G_c(s)G(s)H(s) = 174.3864 \left(\frac{s + 3}{s + 19.1652} \right)^2 \frac{1}{s^2} \frac{1}{0.1s + 1}$$

A root-locus plot for the compensated system is shown in Figure 7-52(a). Notice that there is no closed-loop zero near the origin. An expanded view of the root-locus plot near the origin is shown in Figure 7-52(b).

The closed-loop transfer function becomes

$$\frac{C(s)}{R(s)} = \frac{174.3864(s + 3)^2(0.1s + 1)}{(s + 19.1652)^2 s^2 (0.1s + 1) + 174.3864(s + 3)^2}$$

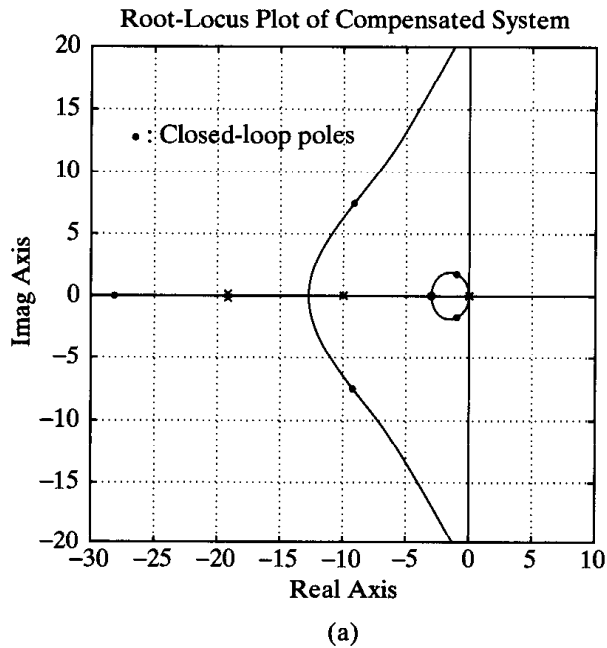


Figure 7-52

(a) Root-locus plot of compensated system; (b) root-locus plot near the origin.

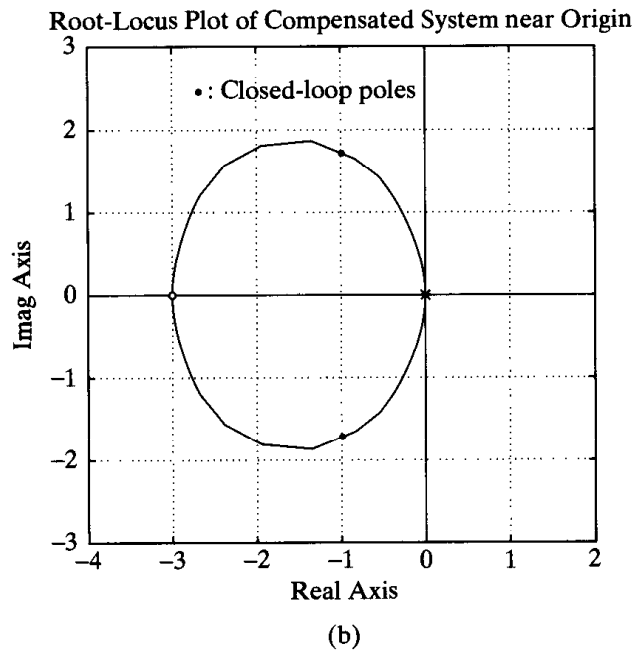


Figure 7-52
(Continued)

The closed-loop poles are found as follows:

$$s = -1 \pm j1.73205$$

$$s = -9.1847 \pm j7.4814$$

$$s = -27.9606$$

Figures 7-53(a) and (b) show the unit-step response and unit-ramp response of the compensated system. The unit-step response curve is reasonable and the unit-ramp response looks acceptable. Notice that in the unit-ramp response the output leads the input by a small amount. This is because the system has a feedback transfer function $1/(0.1s + 1)$. If the feedback signal versus t is plotted, together with the unit-ramp input, the former will not lead the input ramp at steady state. See Figure 7-53(c).

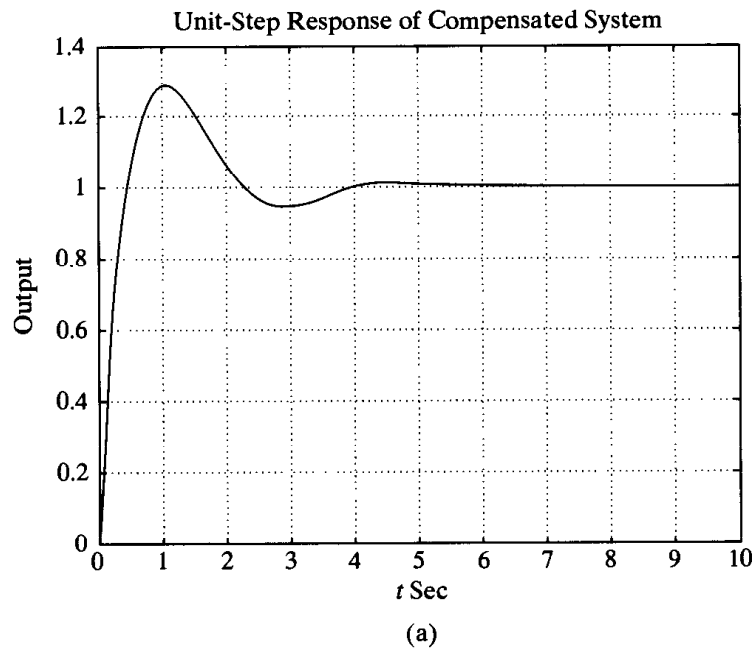


Figure 7-53
(a) Unit-step response of the compensated system;
(b) unit-ramp response of the compensated system;
(c) a plot of feedback signal versus t in the unit-ramp response.

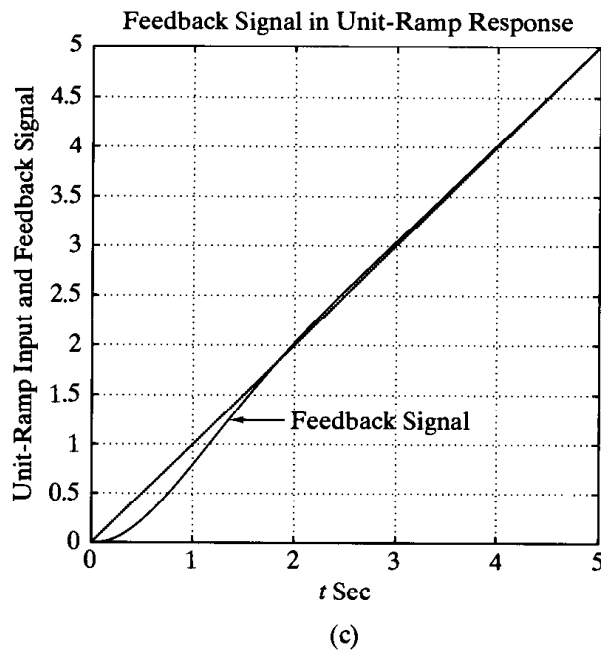
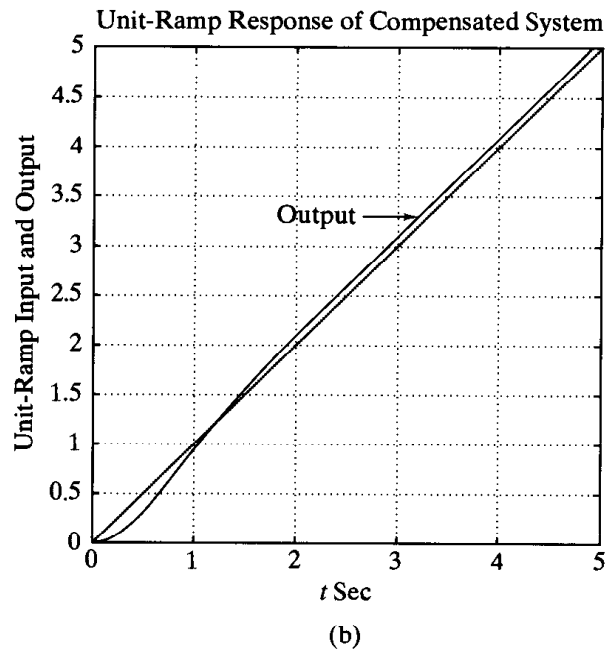


Figure 7-53
(Continued)

PROBLEMS

B-7-1. Consider the mechanical system shown in Figure 7-54. It consists of a spring and two dashpots. Obtain the transfer function of the system. The displacement x_i is the input and displacement x_o is the output. Is this system a mechanical lead network or lag network?

B-7-2. Obtain the transfer function of the electrical network shown in Figure 7-55. Show that it is a lag network.

B-7-3. Consider the system shown in Figure 7-56. Plot the root loci for the system. Determine the value of K such that the damping ratio ζ of the dominant closed-loop poles is 0.5. Then determine all closed-loop poles. Plot the unit-step response curve with MATLAB.

B-7-4. Determine the values of K , T_1 , and T_2 of the system shown in Figure 7-57 so that the dominant closed-loop

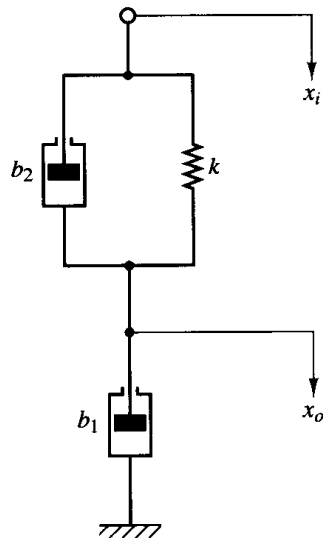


Figure 7-54
Mechanical system.

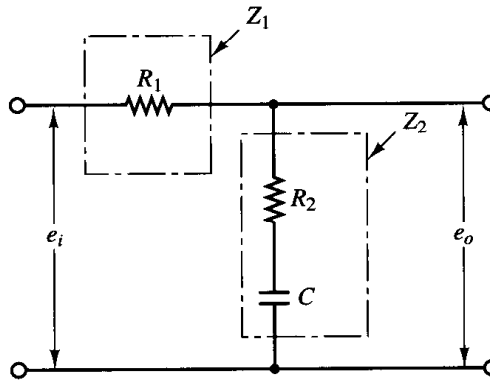


Figure 7-55
Electrical network.

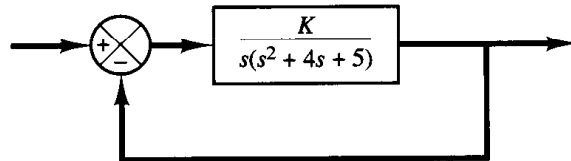


Figure 7-56
Control system.

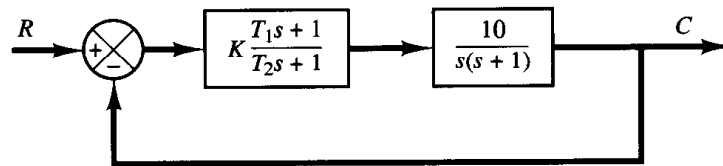


Figure 7-57
Control system.

poles have the damping ratio $\zeta = 0.5$ and the undamped natural frequency $\omega_n = 3$ rad/sec.

B-7-5. Consider the system shown in Figure 7-58, which involves velocity feedback. Determine the values of the amplifier gain K and the velocity feedback gain K_h so that the following specifications are satisfied:

1. Damping ratio of the closed-loop poles is 0.5
2. Settling time ≤ 2 sec
3. Static velocity error constant $K_v \geq 50 \text{ sec}^{-1}$
4. $0 < K_h < 1$

B-7-6. Consider the system shown in Figure 7-59. Design a lead compensator such that the dominant closed-loop poles

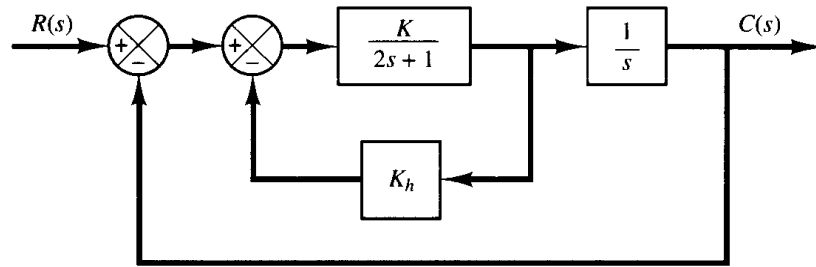


Figure 7-58
Control system.

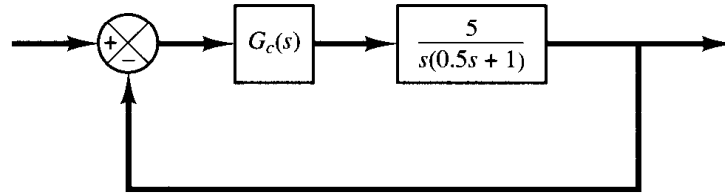


Figure 7-59
Control system.

are located at $s = -2 \pm j2\sqrt{3}$. Plot the unit-step response curve of the designed system with MATLAB.

B-7-7. Consider the system shown in Figure 7-60. Design a compensator such that the dominant closed-loop poles are located at $s = -1 \pm j1$.

B-7-8. Referring to the system shown in Figure 7-61, design a compensator such that the static velocity error constant K_v is 20 sec^{-1} without appreciably changing the original location ($s = -2 \pm j2\sqrt{3}$) of a pair of the complex-conjugate closed-loop poles.

B-7-9. Consider the angular-positional system shown in Figure 7-62. The dominant closed-loop poles are located at $s = -3.60 \pm j4.80$. The damping ratio ζ of the dominant closed-loop poles is 0.6. The static velocity error constant K_v is 4.1 sec^{-1} , which means that for a ramp input of $360^\circ/\text{sec}$ the steady-state error in following the ramp input is

$$e_v = \frac{\theta_i}{K_v} = \frac{360^\circ/\text{sec}}{4.1 \text{ sec}^{-1}} = 87.8^\circ$$

It is desired to decrease e_v to one-tenth of the present value, or to increase the value of the static velocity error constant K_v to 41 sec^{-1} . It is also desired to keep the damping ratio ζ of the dominant closed-loop poles at 0.6. A small change in the undamped natural frequency ω_n of the dominant closed-loop poles is permissible. Design a suitable lag compensator to increase the static velocity error constant as desired.

B-7-10. Consider the control system shown in Figure 7-63. Design a compensator such that the dominant closed-loop poles are located at $s = -2 \pm j2\sqrt{3}$ and the static velocity error constant K_v is 50 sec^{-1} .

B-7-11. Consider the same system as considered in Problem A-7-10. It is desired to design a PID controller $G_c(s)$

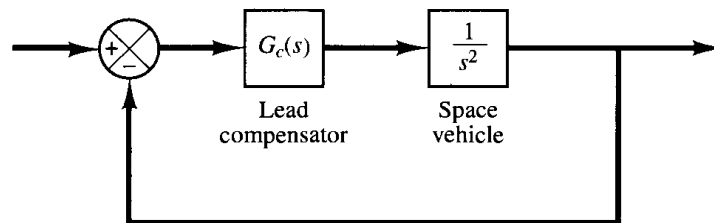


Figure 7-60
Control system.

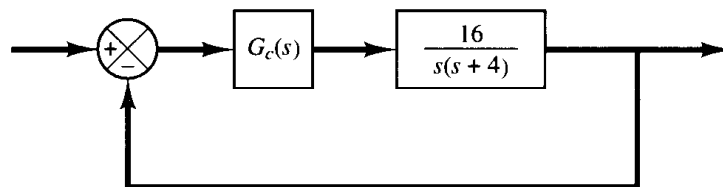


Figure 7-61
Control system.

Figure 7-62
Angular-positional
system.

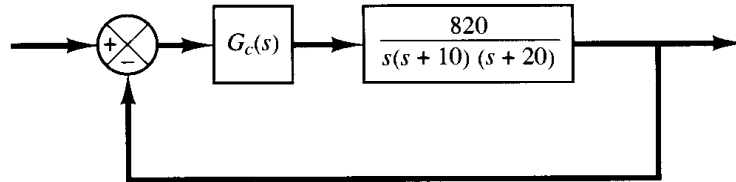
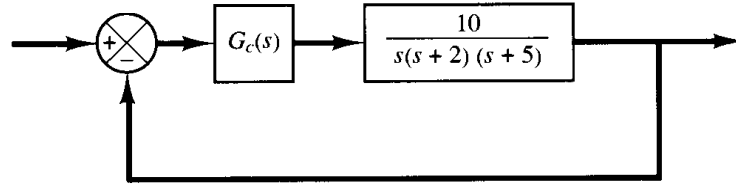


Figure 7-63
Control system.



such that the dominant closed-loop poles are located at $s = -1 \pm j\sqrt{3}$. For the PID controller, choose $a = 0.5$ (instead of $a = 1$ as discussed in Problem A-7-10) and then determine the values of K and b . Sketch the root-locus plot for the designed system. Also, obtain the unit-step response curve with MATLAB.

B-7-12. Consider the control system shown in Figure 7-64. The plant is critically stable in the sense that oscillations will continue indefinitely. Design a suitable compensator such that the unit-step response will exhibit maxi-

mum overshoot of less than 40% and settling time of 5 sec or less.

B-7-13. Consider the control system shown in Figure 7-65. Design a compensator such that the unit-step response curve will exhibit maximum overshoot of 30% or less and settling time of 3 sec or less.

B-7-14. Consider the control system shown in Figure 7-66. Design a compensator such that the unit-step response curve will exhibit maximum overshoot of 25% or less and settling time of 5 sec or less.

Figure 7-64
Control system.

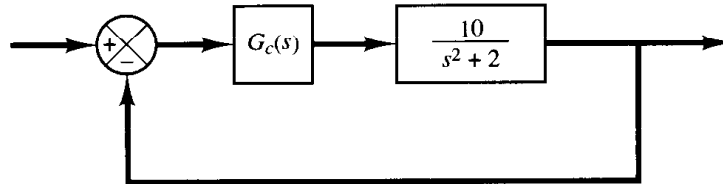


Figure 7-65
Control system.

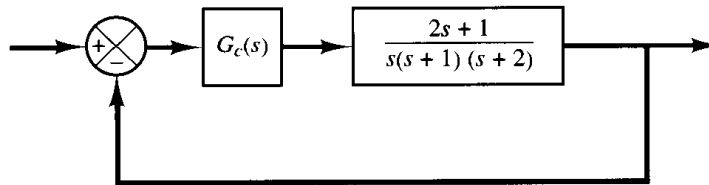
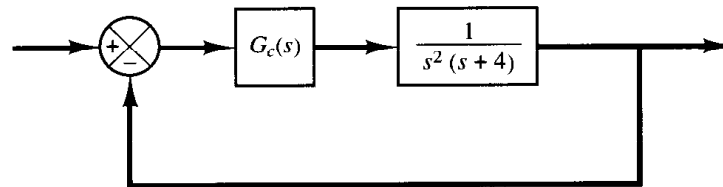


Figure 7-66
Control system.



8

Frequency-Response Analysis

8-1 INTRODUCTION

By the term *frequency response*, we mean the steady-state response of a system to a sinusoidal input. In frequency-response methods, we vary the frequency of the input signal over a certain range and study the resulting response.

The Nyquist stability criterion enables us to investigate both the absolute and relative stabilities of linear closed-loop systems from a knowledge of their open-loop frequency-response characteristics. An advantage of the frequency-response approach is that frequency-response tests are, in general, simple and can be made accurately by use of readily available sinusoidal signal generators and precise measurement equipment. Often the transfer functions of complicated components can be determined experimentally by frequency-response tests. In addition, the frequency-response approach has the advantages that a system may be designed so that the effects of undesirable noise are negligible and that such analysis and design can be extended to certain nonlinear control systems.

Although the frequency response of a control system presents a qualitative picture of the transient response, the correlation between frequency and transient responses is indirect, except for the case of second-order systems. In designing a closed-loop system, we adjust the frequency-response characteristic of the open-loop transfer function by using several design criteria in order to obtain acceptable transient-response characteristics for the system.

Steady-state output to sinusoidal input. Consider the linear time-invariant system shown in Figure 8-1. For this system

$$\frac{Y(s)}{X(s)} = G(s)$$

The input $x(t)$ is sinusoidal and is given by

$$x(t) = X \sin \omega t$$

As presented in Chapter 5, if the system is stable, then the output $y(t)$ can be given by

$$y(t) = Y \sin(\omega t + \phi)$$

where

$$Y = X|G(j\omega)|$$

and

$$\phi = \angle G(j\omega) = \tan^{-1} \left[\frac{\text{imaginary part of } G(j\omega)}{\text{real part of } G(j\omega)} \right]$$

A stable linear time-invariant system subjected to a sinusoidal input will, at steady state, have a sinusoidal output of the same frequency as the input. But the amplitude and phase of the output will, in general, be different from those of the input. In fact, the amplitude of the output is given by the product of that of the input and $|G(j\omega)|$, while the phase angle differs from that of the input by the amount $\phi = \angle G(j\omega)$. An example of input and output sinusoidal signals is shown in Figure 8–2.

Note that for sinusoidal inputs

$$|G(j\omega)| = \left| \frac{Y(j\omega)}{X(j\omega)} \right| = \frac{\text{amplitude ratio of the output sinusoid to the input sinusoid}}{\text{input sinusoid}}$$

$$\angle G(j\omega) = \angle \frac{Y(j\omega)}{X(j\omega)} = \frac{\text{phase shift of the output sinusoid with respect to the input sinusoid}}{\text{the input sinusoid}}$$

Hence, the response characteristics of a system to a sinusoidal input can be obtained directly from

$$\frac{Y(j\omega)}{X(j\omega)} = G(j\omega)$$

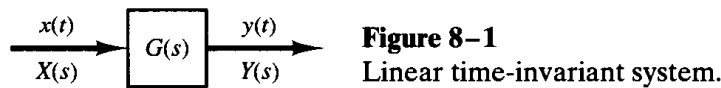


Figure 8–1
Linear time-invariant system.

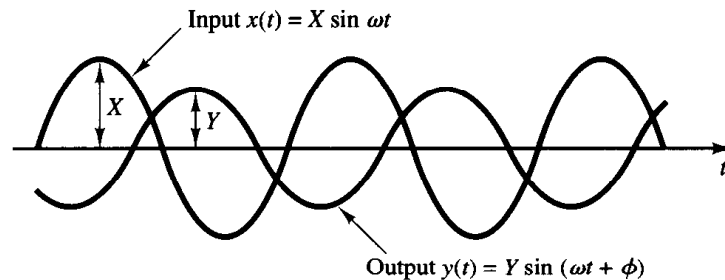


Figure 8–2
Input and output
sinusoidal signals.

The sinusoidal transfer function $G(j\omega)$, the ratio of $Y(j\omega)$ to $X(j\omega)$, is a complex quantity and can be represented by the magnitude and phase angle with frequency as a parameter. (A negative phase angle is called *phase lag*, and a positive phase angle is called *phase lead*.) The sinusoidal transfer function of any linear system is obtained by substituting $j\omega$ for s in the transfer function of the system.

Presenting frequency-response characteristics in graphical forms. The sinusoidal transfer function, a complex function of the frequency ω , is characterized by its magnitude and phase angle, with frequency as the parameter. There are three commonly used representations of sinusoidal transfer functions:

1. Bode diagram or logarithmic plot
2. Nyquist plot or polar plot
3. Log-magnitude versus phase plot

We shall discuss these representations in detail in this chapter. We shall also discuss the MATLAB approach to obtain Bode diagrams and Nyquist plots.

Outline of the chapter. Section 8–1 has presented introductory material on the frequency response. Section 8–2 presents Bode diagrams of various transfer-function systems. Section 8–3 discusses a computational approach to obtain Bode diagrams with MATLAB, Section 8–4 treats polar plots of sinusoidal transfer functions, and Section 8–5 discusses drawing Nyquist plots with MATLAB. Section 8–6 briefly presents log-magnitude versus phase plots. Section 8–7 gives a detailed account of Nyquist stability criterion, Section 8–8 discusses the stability analysis of closed-loop systems using the Nyquist stability criterion, and Section 8–9 treats the relative stability analysis of closed-loop systems. Measures of relative stability such as phase margin and gain margin are introduced here. The correlation between the transient response and frequency response is also discussed. Section 8–10 presents a method for obtaining the closed-loop frequency response from the open-loop frequency response by use of the M and N circles. Use of the Nichols chart is also discussed for obtaining the closed-loop frequency response. Finally, Section 8–11 deals with the determination of the transfer function based on an experimental Bode diagram.

8–2 BODE DIAGRAMS

Bode diagrams or logarithmic plots. A sinusoidal transfer function may be represented by two separate plots, one giving the magnitude versus frequency and the other the phase angle (in degrees) versus frequency. A Bode diagram consists of two graphs: One is a plot of the logarithm of the magnitude of a sinusoidal transfer function; the other is a plot of the phase angle; both are plotted against the frequency in logarithmic scale.

The standard representation of the logarithmic magnitude of $G(j\omega)$ is $20 \log |G(j\omega)|$, where the base of the logarithm is 10. The unit used in this representation of the magnitude is the decibel, usually abbreviated dB. In the logarithmic representation, the curves are drawn on semilog paper, using the log scale for frequency and the linear scale

for either magnitude (but in decibels) or phase angle (in degrees). (The frequency range of interest determines the number of logarithmic cycles required on the abscissa.)

The main advantage of using the Bode diagram is that multiplication of magnitudes can be converted into addition. Furthermore, a simple method for sketching an approximate log-magnitude curve is available. It is based on asymptotic approximations. Such approximation by straight-line asymptotes is sufficient if only rough information on the frequency-response characteristics is needed. Should exact curve be desired, corrections can be made easily to these basic asymptotic ones. The phase-angle curves can be drawn easily if a template for the phase-angle curve of $1 + j\omega$ is available. Expanding the low-frequency range by use of a logarithmic scale for the frequency is very advantageous since characteristics at low frequencies are most important in practical systems. Although it is not possible to plot the curves right down to zero frequency because of the logarithmic frequency ($\log 0 = -\infty$), this does not create a serious problem.

Note that the experimental determination of a transfer function can be made simple if frequency-response data are presented in the form of a Bode diagram.

Basic factors of $G(j\omega)H(j\omega)$. As stated earlier, the main advantage in using the logarithmic plot is the relative ease of plotting frequency-response curves. The basic factors that very frequently occur in an arbitrary transfer function $G(j\omega)H(j\omega)$ are

1. Gain K
2. Integral and derivative factors $(j\omega)^{\mp 1}$
3. First-order factors $(1 + j\omega T)^{\mp 1}$
4. Quadratic factors $[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]^{\mp 1}$

Once we become familiar with the logarithmic plots of these basic factors, it is possible to utilize them in constructing a composite logarithmic plot for any general form of $G(j\omega)H(j\omega)$ by sketching the curves for each factor and adding individual curves graphically, because adding the logarithms of the gains corresponds to multiplying them together.

The process of obtaining the logarithmic plot can be further simplified by using asymptotic approximations to the curves for each factor. (If necessary, corrections can be made easily to an approximate plot to obtain an accurate one.)

The gain K . A number greater than unity has a positive value in decibels, while a number smaller than unity has a negative value. The log-magnitude curve for a constant gain K is a horizontal straight line at the magnitude of $20 \log K$ decibels. The phase angle of the gain K is zero. The effect of varying the gain K in the transfer function is that it raises or lowers the log-magnitude curve of the transfer function by the corresponding constant amount, but it has no effect on the phase curve.

A number-decibel conversion line is given in Figure 8-3. The decibel value of any number can be obtained from this line. As a number increases by a factor of 10, the corresponding decibel value increases by a factor of 20. This may be seen from the following:

$$20 \log(K \times 10) = 20 \log K + 20$$

Similarly,

$$20 \log(K \times 10^n) = 20 \log K + 20n$$

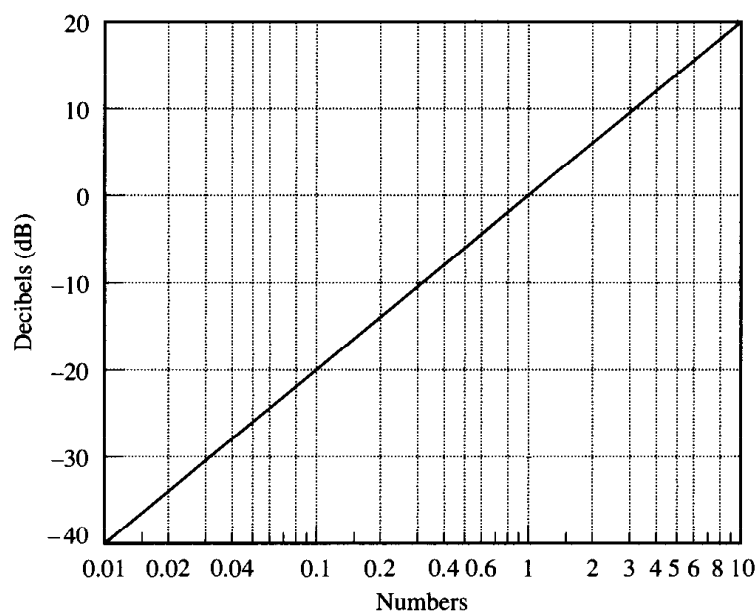


Figure 8-3
Number-decibel
conversion line.

Note that, when expressed in decibels, the reciprocal of a number differs from its value only in sign; that is, for the number K ,

$$20 \log K = -20 \log \frac{1}{K}$$

Integral and derivative factors $(j\omega)^{\pm 1}$. The logarithmic magnitude of $1/j\omega$ in decibels is

$$20 \log \left| \frac{1}{j\omega} \right| = -20 \log \omega \text{ dB}$$

The phase angle of $1/j\omega$ is constant and equal to -90° .

In Bode diagrams, frequency ratios are expressed in terms of octaves or decades. An octave is a frequency band from ω_1 to $2\omega_1$, where ω_1 is any frequency value. A decade is a frequency band from ω_1 to $10\omega_1$, where again ω_1 is any frequency. (On the logarithmic scale of semilog paper, any given frequency ratio can be represented by the same horizontal distance. For example, the horizontal distance from $\omega = 1$ to $\omega = 10$ is equal to that from $\omega = 3$ to $\omega = 30$.)

If the log magnitude $-20 \log \omega$ dB is plotted against ω on a logarithmic scale, it is a straight line. To draw this straight line, we need to locate one point (0 dB, $\omega = 1$) on it. Since

$$(-20 \log 10\omega) \text{ dB} = (-20 \log \omega - 20) \text{ dB}$$

the slope of the line is -20 dB/decade (or -6 dB/octave).

Similarly, the log magnitude of $j\omega$ in decibels is

$$20 \log |j\omega| = 20 \log \omega \text{ dB}$$

The phase angle of $j\omega$ is constant and equal to 90° . The log-magnitude curve is a straight line with a slope of 20 dB/decade. Figures 8-4 (a) and (b) show frequency-response

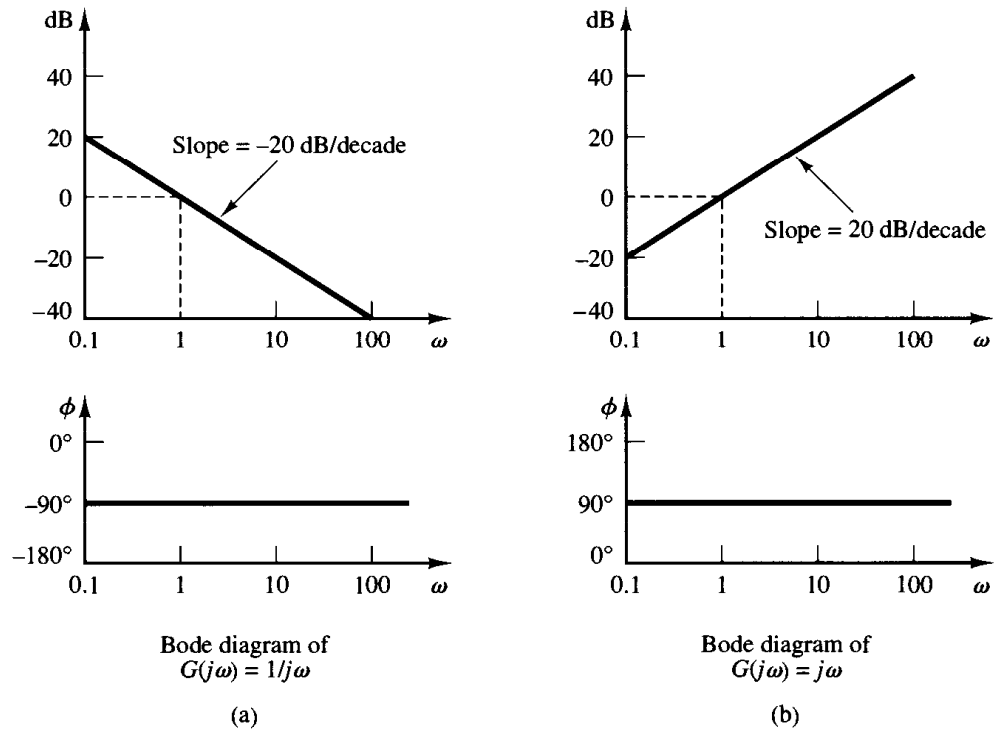


Figure 8-4

(a) Bode diagram of $G(j\omega) = 1/j\omega$;
 (b) Bode diagram of $G(j\omega) = j\omega$.

curves for $1/j\omega$ and $j\omega$, respectively. We can clearly see that the differences in the frequency responses of the factors $1/j\omega$ and $j\omega$ lie in the signs of the slopes of the log-magnitude curves and in the signs of the phase angles. Both log magnitudes become equal to 0 dB at $\omega = 1$.

If the transfer function contains the factor $(1/j\omega)^n$ or $(j\omega)^n$, the log magnitude becomes, respectively,

$$20 \log \left| \frac{1}{(j\omega)^n} \right| = -n \times 20 \log |j\omega| = -20n \log \omega \text{ dB}$$

or

$$20 \log |(j\omega)^n| = n \times 20 \log |j\omega| = 20n \log \omega \text{ dB}$$

The slopes of the log-magnitude curves for the factors $(1/j\omega)^n$ and $(j\omega)^n$ are thus $-20n$ dB/decade and $20n$ dB/decade, respectively. The phase angle of $(1/j\omega)^n$ is equal to $-90^\circ \times n$ over the entire frequency range, while that of $(j\omega)^n$ is equal to $90^\circ \times n$ over the entire frequency range. The magnitude curves will pass through the point (0 dB, $\omega = 1$).

First-order factors $(1 + j\omega T)^{\mp 1}$. The log magnitude of the first-order factor $1/(1 + j\omega T)$ is

$$20 \log \left| \frac{1}{1 + j\omega T} \right| = -20 \log \sqrt{1 + \omega^2 T^2} \text{ dB}$$

For low frequencies, such that $\omega \ll 1/T$, the log magnitude may be approximated by

$$-20 \log \sqrt{1 + \omega^2 T^2} \doteq -20 \log 1 = 0 \text{ dB}$$

Thus, the log-magnitude curve at low frequencies is the constant 0-dB line. For high frequencies, such that $\omega \gg 1/T$,

$$-20 \log \sqrt{1 + \omega^2 T^2} \doteq -20 \log \omega T \text{ dB}$$

This is an approximate expression for the high-frequency range. At $\omega = 1/T$, the log magnitude equals 0 dB; at $\omega = 10/T$, the log magnitude is -20 dB. Thus, the value of $-20 \log \omega T$ dB decreases by 20 dB for every decade of ω . For $\omega \gg 1/T$, the log-magnitude curve is thus a straight line with a slope of -20 dB/decade (or -6 dB/octave).

Our analysis shows that the logarithmic representation of the frequency-response curve of the factor $1/(1 + j\omega T)$ can be approximated by two straight-line asymptotes, one a straight line at 0 dB for the frequency range $0 < \omega < 1/T$ and the other a straight line with slope -20 dB/decade (or -6 dB/octave) for the frequency range $1/T < \omega < \infty$. The exact log-magnitude curve, the asymptotes, and the exact phase-angle curve are shown in Figure 8-5.

The frequency at which the two asymptotes meet is called the *corner frequency* or *break frequency*. For the factor $1/(1 + j\omega T)$, the frequency $\omega = 1/T$ is the corner frequency since at $\omega = 1/T$ the two asymptotes have the same value. (The low-frequency asymptotic expression at $\omega = 1/T$ is $20 \log 1 \text{ dB} = 0 \text{ dB}$, and the high-frequency asymptotic expression at $\omega = 1/T$ is also $20 \log 1 \text{ dB} = 0 \text{ dB}$.) The corner frequency divides the frequency-response curve into two regions, a curve for the low-frequency region and a curve for the high-frequency region. The corner frequency is very important in sketching logarithmic frequency-response curves.

The exact phase angle ϕ of the factor $1/(1 + j\omega T)$ is

$$\phi = -\tan^{-1} \omega T$$

At zero frequency, the phase angle is 0° . At the corner frequency, the phase angle is

$$\phi = -\tan^{-1} \frac{T}{T} = -\tan^{-1} 1 = -45^\circ$$

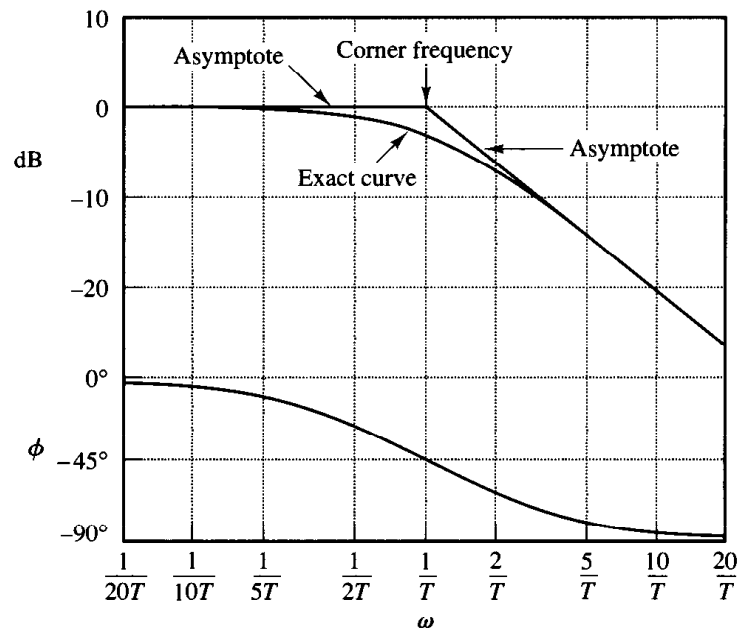


Figure 8-5
Log-magnitude curve
together with the as-
ymptotes and phase
angle curve of
 $1/(1 + j\omega T)$.

At infinity, the phase angle becomes -90° . Since the phase angle is given by an inverse-tangent function, the phase angle is skew symmetric about the inflection point at $\phi = -45^\circ$.

The error in the magnitude curve caused by the use of asymptotes can be calculated. The maximum error occurs at the corner frequency and is approximately equal to -3 dB since

$$-20 \log \sqrt{1 + 1} + 20 \log 1 = -10 \log 2 = -3.03 \text{ dB}$$

The error at the frequency one octave below the corner frequency, that is, at $\omega = 1/(2T)$, is

$$-20 \log \sqrt{\frac{1}{4} + 1} + 20 \log 1 = -20 \log \frac{\sqrt{5}}{2} = -0.97 \text{ dB}$$

The error at the frequency one octave above the corner frequency, that is, at $\omega = 2/T$, is

$$-20 \log \sqrt{2^2 + 1} + 20 \log 2 = -20 \log \frac{\sqrt{5}}{2} = -0.97 \text{ dB}$$

Thus, the error at one octave below or above the corner frequency is approximately equal to -1 dB. Similarly, the error at one decade below or above the corner frequency is approximately -0.04 dB. The error in decibels involved in using the asymptotic expression for the frequency-response curve of $1/(1 + j\omega T)$ is shown in Figure 8-6. The error is symmetric with respect to the corner frequency.

Since the asymptotes are quite easy to draw and are sufficiently close to the exact curve, the use of such approximations in drawing Bode diagrams is convenient in establishing the general nature of the frequency-response characteristics quickly with a minimum amount of calculation and may be used for most preliminary design work. If accurate frequency-response curves are desired, corrections may easily be made by referring to the curve given in Figure 8-6. In practice, an accurate frequency-response curve can be drawn by introducing a correction of 3 dB at the corner frequency and a correction of 1 dB at points one octave below and above the corner frequency and then connecting these points by a smooth curve.

Note that varying the time constant T shifts the corner frequency to the left or to the right, but the shapes of the log-magnitude and the phase-angle curves remain the same.

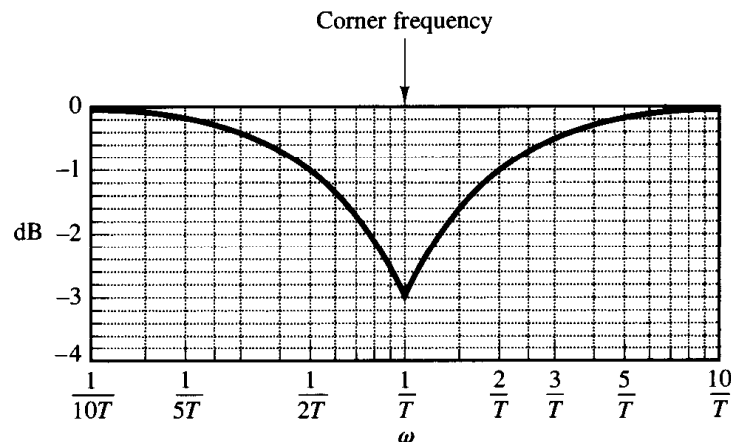


Figure 8-6
Log-magnitude error
in the asymptotic
expression of the
frequency-response
curve of $1/(1 + j\omega T)$.

The transfer function $1/(1 + j\omega T)$ has the characteristics of a low-pass filter. For frequencies above $\omega = 1/T$, the log magnitude falls off rapidly toward $-\infty$. This is essentially due to the presence of the time constant. In the low-pass filter, the output can follow a sinusoidal input faithfully at low frequencies. But as the input frequency is increased, the output cannot follow the input because a certain amount of time is required for the system to build up in magnitude. Thus, at high frequencies, the amplitude of the output approaches zero and the phase angle of the output approaches -90° . Therefore, if the input function contains many harmonics, then the low-frequency components are reproduced faithfully at the output, while the high-frequency components are attenuated in amplitude and shifted in phase. Thus, a first-order element yields exact, or almost exact, duplication only for constant or slowly varying phenomena.

An advantage of the Bode diagram is that for reciprocal factors, for example, the factor $1 + j\omega T$, the log-magnitude and the phase-angle curves need only be changed in sign. Since

$$20 \log |1 + j\omega T| = -20 \log \left| \frac{1}{1 + j\omega T} \right|$$

$$\angle 1 + j\omega T = \tan^{-1} \omega T = - \angle \frac{1}{1 + j\omega T}$$

the corner frequency is the same for both cases. The slope of the high-frequency asymptote of $1 + j\omega T$ is 20 dB/decade, and the phase angle varies from 0° to 90° as the frequency ω is increased from zero to infinity. The log-magnitude curve together with the asymptotes and the phase-angle curve for the factor $1 + j\omega T$ are shown in Figure 8-7.

The shapes of phase-angle curves are the same for any factor of the form $(1 + j\omega T)^{\pm 1}$. Hence, it is convenient to have a template for the phase-angle curve on cardboard. Then such a template may be used repeatedly for constructing phase-angle

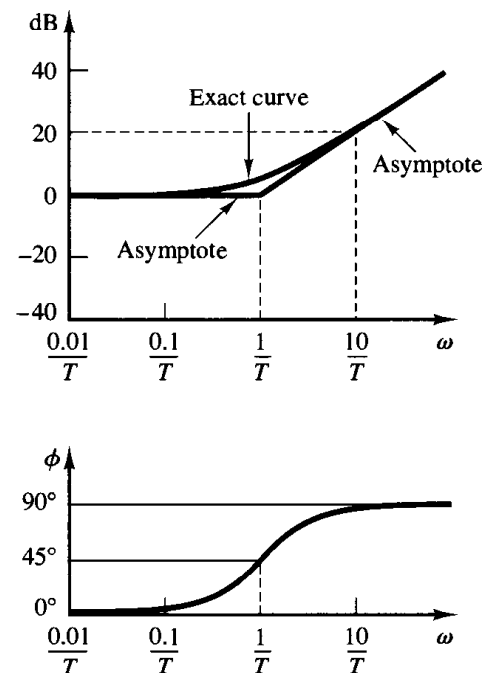


Figure 8-7
Log-magnitude curve together with the asymptotes and phase-angle curve for $1 + j\omega T$.

curves for any function of the form $(1 + j\omega T)^{\mp 1}$. If such a template is not available, we have to locate several points on the curve. The phase angles of $(1 + j\omega T)^{\mp 1}$ are

$$\begin{array}{lll} \mp 45^\circ & \text{at} & \omega = \frac{1}{T} \\ \mp 26.6^\circ & \text{at} & \omega = \frac{1}{2T} \\ \mp 5.7^\circ & \text{at} & \omega = \frac{1}{10T} \\ \mp 63.4^\circ & \text{at} & \omega = \frac{2}{T} \\ \mp 84.3^\circ & \text{at} & \omega = \frac{10}{T} \end{array}$$

For the case where a given transfer function involves terms like $(1 + j\omega T)^{\mp n}$, a similar asymptotic construction may be made. The corner frequency is still at $\omega = 1/T$, and the asymptotes are straight lines. The low-frequency asymptote is a horizontal straight line at 0 dB, while the high-frequency asymptote has the slope of $-20n$ dB/decade or $20n$ dB/decade. The error involved in the asymptotic expressions is n times that for $(1 + j\omega T)^{\mp 1}$. The phase angle is n times that of $(1 + j\omega T)^{\mp 1}$ at each frequency point.

Quadratic factors $[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]^{\mp 1}$. Control systems often possess quadratic factors of the form

$$\frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2} \quad (8-1)$$

If $\zeta > 1$, this quadratic factor can be expressed as a product of two first-order factors with real poles. If $0 < \zeta < 1$, this quadratic factor is the product of two complex-conjugate factors. Asymptotic approximations to the frequency-response curves are not accurate for a factor with low values of ζ . This is because the magnitude and phase of the quadratic factor depend on both the corner frequency and the damping ratio ζ .

The asymptotic frequency-response curve may be obtained as follows: Since

$$20 \log \left| \frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2} \right| = -20 \log \sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta \frac{\omega}{\omega_n}\right)^2}$$

for low frequencies such that $\omega \ll \omega_n$, the log magnitude becomes

$$-20 \log 1 = 0 \text{ dB}$$

The low-frequency asymptote is thus a horizontal line at 0 dB. For high frequencies such that $\omega \gg \omega_n$, the log magnitude becomes

$$-20 \log \frac{\omega^2}{\omega_n^2} = -40 \log \frac{\omega}{\omega_n} \text{ dB}$$

The equation for the high-frequency asymptote is a straight line having the slope -40 dB/decade since

$$-40 \log \frac{10\omega}{\omega_n} = -40 - 40 \log \frac{\omega}{\omega_n}$$

The high-frequency asymptote intersects the low-frequency one at $\omega = \omega_n$ since at this frequency

$$-40 \log \frac{\omega_n}{\omega_n} = -40 \log 1 = 0 \text{ dB}$$

This frequency, ω_n , is the corner frequency for the quadratic factor considered.

The two asymptotes just derived are independent of the value of ζ . Near the frequency $\omega = \omega_n$, a resonant peak occurs, as may be expected from (8-1). The damping ratio ζ determines the magnitude of this resonant peak. Errors obviously exist in the approximation by straight-line asymptotes. The magnitude of the error depends on the value of ζ . It is large for small values of ζ . Figure 8-8 shows the exact log-magnitude curves together with the straight-line asymptotes and the exact phase-angle curves for the quadratic factor given by (8-1) with several values of ζ . If corrections are desired in the asymptotic curves, the necessary amounts of correction at a sufficient number of frequency points may be obtained from Figure 8-8.

The phase angle of the quadratic factor $[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]^{-1}$ is

$$\phi = \angle \frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2} = -\tan^{-1} \left[\frac{2\zeta \frac{\omega}{\omega_n}}{1 - \left(\frac{\omega}{\omega_n}\right)^2} \right] \quad (8-2)$$

The phase angle is a function of both ω and ζ . At $\omega = 0$, the phase angle equals 0° . At the corner frequency $\omega = \omega_n$, the phase angle is -90° regardless of ζ since

$$\phi = -\tan^{-1} \left(\frac{2\zeta}{0} \right) = -\tan^{-1} \infty = -90^\circ$$

At $\omega = \infty$, the phase angle becomes -180° . The phase-angle curve is skew symmetric about the inflection point, the point where $\phi = -90^\circ$. There are no simple ways to sketch such phase curves. We need to refer to the phase-angle curves shown in Figure 8-8.

The frequency-response curves for the factor

$$1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2$$

can be obtained by merely reversing the sign of the log magnitude and that of the phase angle of the factor

$$\frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2}$$

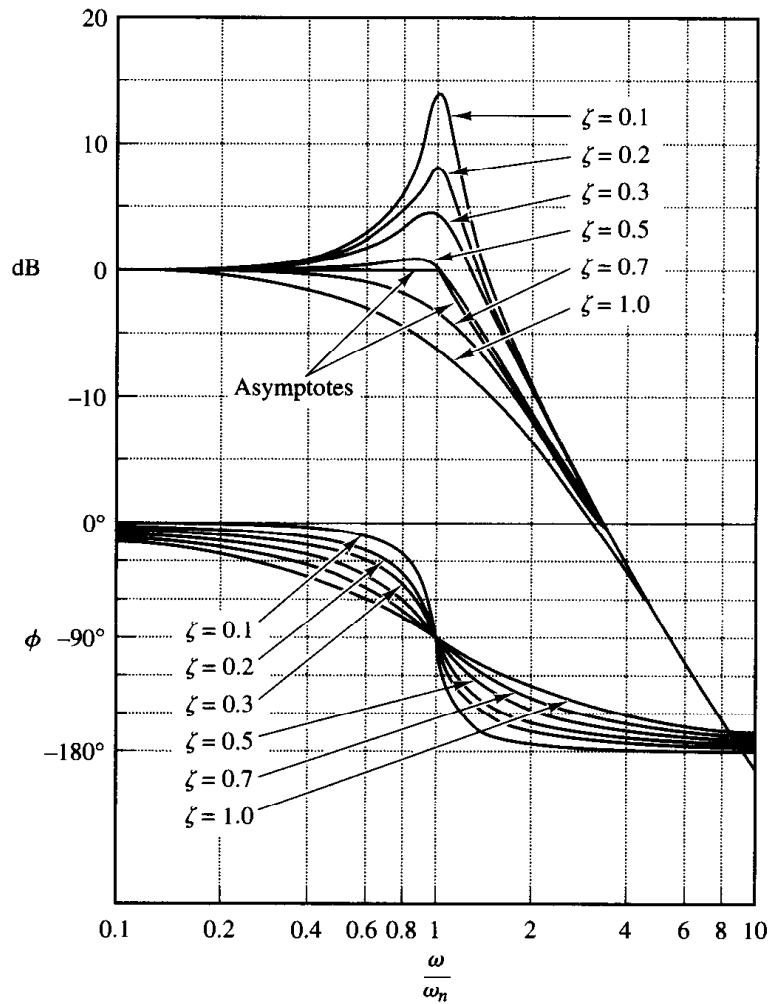


Figure 8-8
Log-magnitude
curves together with
the asymptotes and
phase-angle curves
of the quadratic
transfer function
given by (8-1).

To obtain the frequency-response curves of a given quadratic transfer function, we must first determine the value of the corner frequency ω_n and that of the damping ratio ζ . Then, by using the family of curves given in Figure 8-8, the frequency-response curves can be plotted.

The resonant frequency ω_r and the resonant peak value M_r . The magnitude of

$$G(j\omega) = \frac{1}{1 + 2\zeta \left(j \frac{\omega}{\omega_n} \right) + \left(j \frac{\omega}{\omega_n} \right)^2}$$

is

$$|G(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta \frac{\omega}{\omega_n}\right)^2}} \quad (8-3)$$

If $|G(j\omega)|$ has a peak value at some frequency, this frequency is called the *resonant* frequency. Since the numerator of $|G(j\omega)|$ is constant, a peak value of $|G(j\omega)|$ will occur when

$$g(\omega) = \left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta \frac{\omega}{\omega_n}\right)^2 \quad (8-4)$$

is a minimum. Since Equation (8-4) can be written

$$g(\omega) = \left[\frac{\omega^2 - \omega_n^2(1 - 2\zeta^2)}{\omega_n^2} \right]^2 + 4\zeta^2(1 - \zeta^2) \quad (8-5)$$

the minimum value of $g(\omega)$ occurs at $\omega = \omega_n\sqrt{1 - 2\zeta^2}$. Thus the resonant frequency ω_r is

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}, \quad \text{for } 0 \leq \zeta \leq 0.707 \quad (8-6)$$

As the damping ratio ζ approaches zero, the resonant frequency approaches ω_n . For $0 < \zeta \leq 0.707$, the resonant frequency ω_r is less than the damped natural frequency $\omega_d = \omega_n\sqrt{1 - \zeta^2}$, which is exhibited in the transient response. From Equation (8-6) it can be seen that for $\zeta > 0.707$ there is no resonant peak. The magnitude $|G(j\omega)|$ decreases monotonically with increasing frequency ω . (The magnitude is less than 0 dB for all values of $\omega > 0$. Recall that, for $0.7 < \zeta < 1$, the step response is oscillatory, but the oscillations are well damped and are hardly perceptible.)

The magnitude of the resonant peak M_r can be found by substituting Equation (8-6) into Equation (8-3). For $0 \leq \zeta \leq 0.707$,

$$M_r = |G(j\omega)|_{\max} = |G(j\omega_r)| = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \quad (8-7)$$

For $\zeta > 0.707$,

$$M_r = 1 \quad (8-8)$$

As ζ approaches zero, M_r approaches infinity. This means that if the undamped system is excited at its natural frequency the magnitude of $G(j\omega)$ becomes infinity. The relationship between M_r and ζ is shown in Figure 8-9.

The phase angle of $G(j\omega)$ at the frequency where the resonant peak occurs can be obtained by substituting Equation (8-6) into Equation (8-2). Thus, at the resonant frequency ω_r ,

$$\angle G(j\omega_r) = -\tan^{-1} \frac{\sqrt{1 - 2\zeta^2}}{\zeta} = -90^\circ + \sin^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}$$

General procedure for plotting Bode diagrams. First rewrite the sinusoidal-transfer function $G(j\omega)H(j\omega)$ as a product of basic factors discussed above. Then

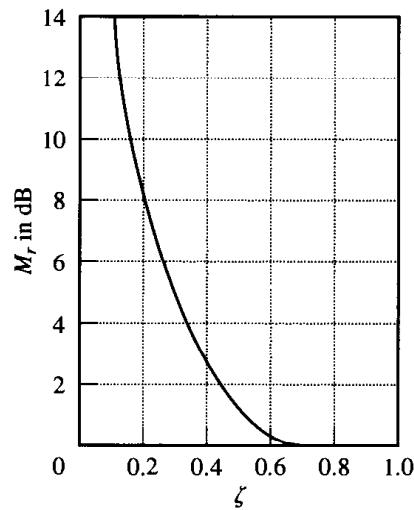


Figure 8-9
 M_r versus ζ curve for the second-order system $1/[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]$.

identify the corner frequencies associated with these basic factors. Finally, draw the asymptotic log-magnitude curves with proper slopes between the corner frequencies. The exact curve, which lies close to the asymptotic curve, can be obtained by adding proper corrections.

The phase-angle curve of $G(j\omega)H(j\omega)$ can be drawn by adding the phase-angle curves of individual factors.

The use of Bode diagrams employing asymptotic approximations requires much less time than other methods that may be used for computing the frequency response of a transfer function. The ease of plotting the frequency-response curves for a given transfer function and the ease of modification of the frequency-response curve as compensation is added are the main reasons why Bode diagrams are very frequently used in practice.

EXAMPLE 8-1

Draw the Bode diagram for the following transfer function:

$$G(j\omega) = \frac{10(j\omega + 3)}{(j\omega)(j\omega + 2)[(j\omega)^2 + j\omega + 2]}$$

Make corrections so that the log-magnitude curve is accurate.

To avoid any possible mistakes in drawing the log-magnitude curve, it is desirable to put $G(j\omega)$ in the following normalized form, where the low-frequency asymptotes for the first-order factors and the second-order factor are the 0-dB line.

$$G(j\omega) = \frac{7.5\left(\frac{j\omega}{3} + 1\right)}{(j\omega)\left(\frac{j\omega}{2} + 1\right)\left[\frac{(j\omega)^2}{2} + \frac{j\omega}{2} + 1\right]}$$

This function is composed of the following factors:

$$7.5, \quad (j\omega)^{-1}, \quad 1 + j\frac{\omega}{3}, \quad \left(1 + j\frac{\omega}{2}\right)^{-1}, \quad \left[1 + j\frac{\omega}{2} + \frac{(j\omega)^2}{2}\right]^{-1}$$

The corner frequencies of the third, fourth, and fifth terms are $\omega = 3$, $\omega = 2$, and $\omega = \sqrt{2}$, respectively. Note that the last term has the damping ratio of 0.3536.

To plot the Bode diagram, the separate asymptotic curves for each of the factors are shown in Figure 8-10. The composite curve is then obtained by adding algebraically the individual curves, also shown in Figure 8-10. Note that when the individual asymptotic curves are added at each frequency the slope of the composite curve is cumulative. Below $\omega = \sqrt{2}$, the plot has the slope of -20 dB/decade. At the first corner frequency $\omega = \sqrt{2}$, the slope changes to -60 dB/decade and continues to the next corner frequency $\omega = 2$, where the slope becomes -80 dB/decade. At the last corner frequency $\omega = 3$, the slope changes to -60 dB/decade.

Once such an approximate log-magnitude curve has been drawn, the actual curve can be obtained by adding corrections at each corner frequency and at frequencies one octave below and above the corner frequencies. For first-order factors $(1 + j\omega T)^{\pm 1}$, the corrections are ± 3 dB at the corner frequency and ± 1 dB at the frequencies one octave below and above the corner frequency. Corrections necessary for the quadratic factor are obtained from Figure 8-8. The exact log-magnitude curve for $G(j\omega)$ is shown by a dashed curve in Figure 8-10.

Note that any change in the slope of the magnitude curve is made only at the corner frequencies of the transfer function $G(j\omega)$. Therefore, instead of drawing individual magnitude curves and adding them up, as shown, we may sketch the magnitude curve without sketching individual curves. We may start drawing the lowest-frequency portion of the straight line (that is, the straight

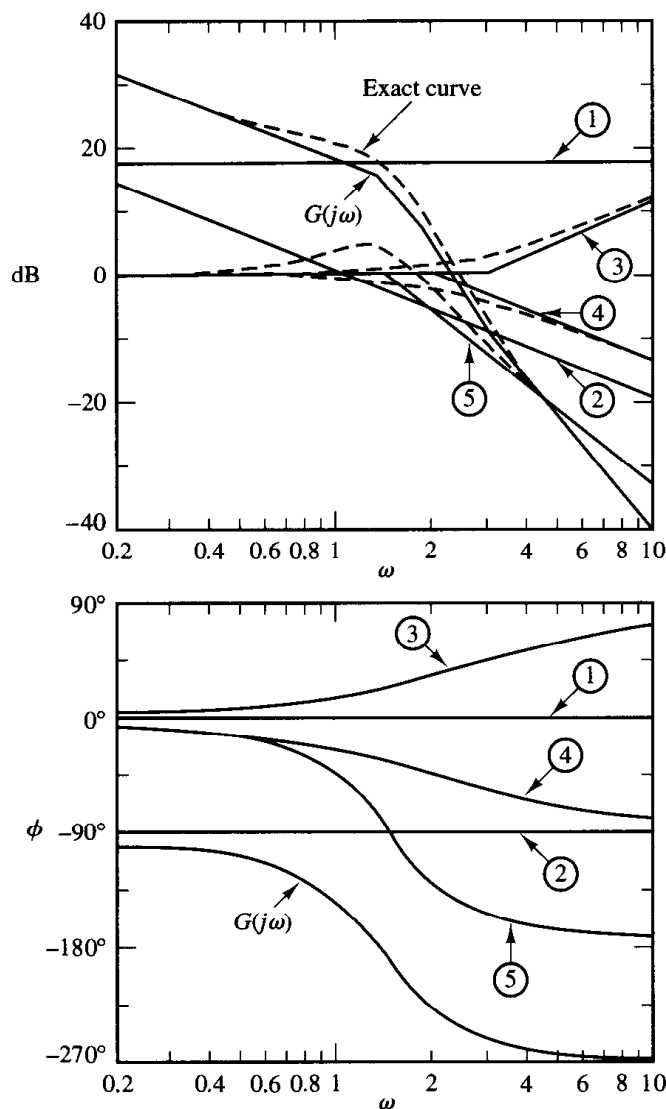


Figure 8-10
Bode diagram of the system considered in Example 8-1.

line with the slope -20 dB/decade for $\omega < \sqrt{2}$. As the frequency is increased, we get the effect of the complex-conjugate poles (quadratic term) at the corner frequency $\omega = \sqrt{2}$. The complex-conjugate poles cause the slopes of the magnitude curve to change from -20 to -60 dB/decade. At the next corner frequency, $\omega = 2$, the effect of the pole is to change the slope to -80 dB/decade. Finally, at the corner frequency $\omega = 3$, the effect of the zero is to change the slope from -80 to -60 dB/decade.

For plotting the complete phase-angle curve, the phase-angle curves for all factors have to be sketched. The algebraic sum of all phase-angle curves provides the complete phase-angle curve, as shown in Figure 8-10.

Minimum-phase systems and nonminimum-phase systems. Transfer functions having neither poles nor zeros in the right-half s plane are minimum-phase transfer functions, whereas those having poles and/or zeros in the right-half s plane are nonminimum-phase transfer functions. Systems with minimum-phase transfer functions are called *minimum-phase* systems, whereas those with nonminimum-phase transfer functions are called *nonminimum-phase* systems.

For systems with the same magnitude characteristic, the range in phase angle of the minimum-phase transfer function is minimum among all such systems, while the range in phase angle of any nonminimum-phase transfer function is greater than this minimum.

It is noted that for a minimum-phase system the transfer function can be uniquely determined from the magnitude curve alone. For a nonminimum-phase system, this is not the case. Multiplying any transfer function by all-pass filters does not alter the magnitude curve, but the phase curve is changed.

Consider as an example the two systems whose sinusoidal transfer functions are, respectively,

$$G_1(j\omega) = \frac{1 + j\omega T}{1 + j\omega T_1}, \quad G_2(j\omega) = \frac{1 - j\omega T}{1 + j\omega T_1} \quad 0 < T < T_1$$

The pole-zero configurations of these systems are shown in Figure 8-11. The two sinusoidal transfer functions have the same magnitude characteristics, but they have different phase-angle characteristics, as shown in Figure 8-12. These two systems differ from each other by the factor

$$G(j\omega) = \frac{1 - j\omega T}{1 + j\omega T}$$

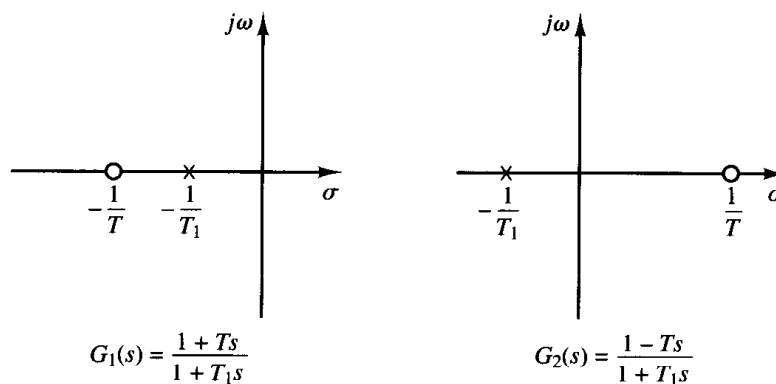
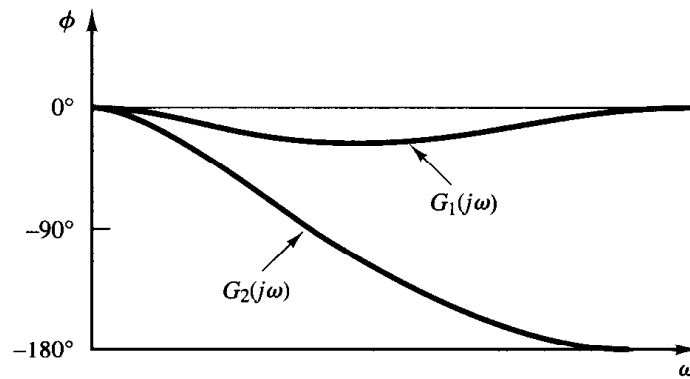


Figure 8-11
Pole-zero configurations of a minimum-phase system $G_1(s)$ and nonminimum-phase system $G_2(s)$.

Figure 8-12
Phase-angle characteristics of the systems $G_1(s)$ and $G_2(s)$ shown in Figure 8-11.



The magnitude of the factor $(1 - j\omega T)/(1 + j\omega T)$ is always unity. But the phase angle equals $-2 \tan^{-1} \omega T$ and varies from 0° to -180° as ω is increased from zero to infinity.

As stated earlier, for a minimum-phase system, the magnitude and phase-angle characteristics are uniquely related. This means that if the magnitude curve of a system is specified over the entire frequency range from zero to infinity, then the phase-angle curve is uniquely determined, and vice versa. This, however, does not hold for a nonminimum-phase system.

Nonminimum-phase situations may arise in two different ways. One is simply when a system includes a nonminimum-phase element or elements. The other situation may arise in the case where a minor loop is unstable.

For a minimum-phase system, the phase angle at $\omega = \infty$ becomes $-90^\circ(q - p)$, where p and q are the degrees of the numerator and denominator polynomials of the transfer function, respectively. For a nonminimum-phase system, the phase angle at $\omega = \infty$ differs from $-90^\circ(q - p)$. In either system, the slope of the log-magnitude curve at $\omega = \infty$ is equal to $-20(q - p)$ dB/decade. It is therefore possible to detect whether the system is minimum phase by examining both the slope of the high-frequency asymptote of the log-magnitude curve and the phase angle at $\omega = \infty$. If the slope of the log-magnitude curve as ω approaches infinity is $-20(q - p)$ dB/decade and the phase angle at $\omega = \infty$ is equal to $-90^\circ(q - p)$, then the system is minimum phase.

Nonminimum-phase systems are slow in response because of their faulty behavior at the start of response. In most practical control systems, excessive phase lag should be carefully avoided. In designing a system, if fast speed of response is of primary importance, we should not use nonminimum-phase components. (A common example of nonminimum-phase elements that may be present in control system is transport lag.)

It is noted that the techniques of frequency-response analysis and design to be presented in this and the next chapter are valid for both minimum-phase and nonminimum-phase systems.

Transport lag. Transport lag is of nonminimum-phase behavior and has an excessive phase lag with no attenuation at high frequencies. Such transport lags normally exist in thermal, hydraulic, and pneumatic systems.

Consider the transport lag given by

$$G(j\omega) = e^{-j\omega T}$$

The magnitude is always equal to unity since

$$|G(j\omega)| = |\cos \omega T - j \sin \omega T| = 1$$

Therefore, the log magnitude of the transport lag $e^{-j\omega T}$ is equal to 0 dB. The phase angle of the transport lag is

$$\begin{aligned} \angle G(j\omega) &= -\omega T \quad (\text{radians}) \\ &= -57.3 \omega T \quad (\text{degrees}) \end{aligned}$$

The phase angle varies linearly with the frequency ω . The phase-angle characteristic of transport lag is shown in Figure 8–13.

EXAMPLE 8–2

Draw the Bode diagram of the following transfer function:

$$G(j\omega) = \frac{e^{-j\omega L}}{1 + j\omega T}$$

The log magnitude is

$$\begin{aligned} 20 \log |G(j\omega)| &= 20 \log |e^{-j\omega L}| + 20 \log \left| \frac{1}{1 + j\omega T} \right| \\ &= 0 + 20 \log \left| \frac{1}{1 + j\omega T} \right| \end{aligned}$$

The phase angle of $G(j\omega)$ is

$$\begin{aligned} \angle G(j\omega) &= \angle e^{-j\omega L} + \angle \frac{1}{1 + j\omega T} \\ &= -\omega L - \tan^{-1} \omega T \end{aligned}$$

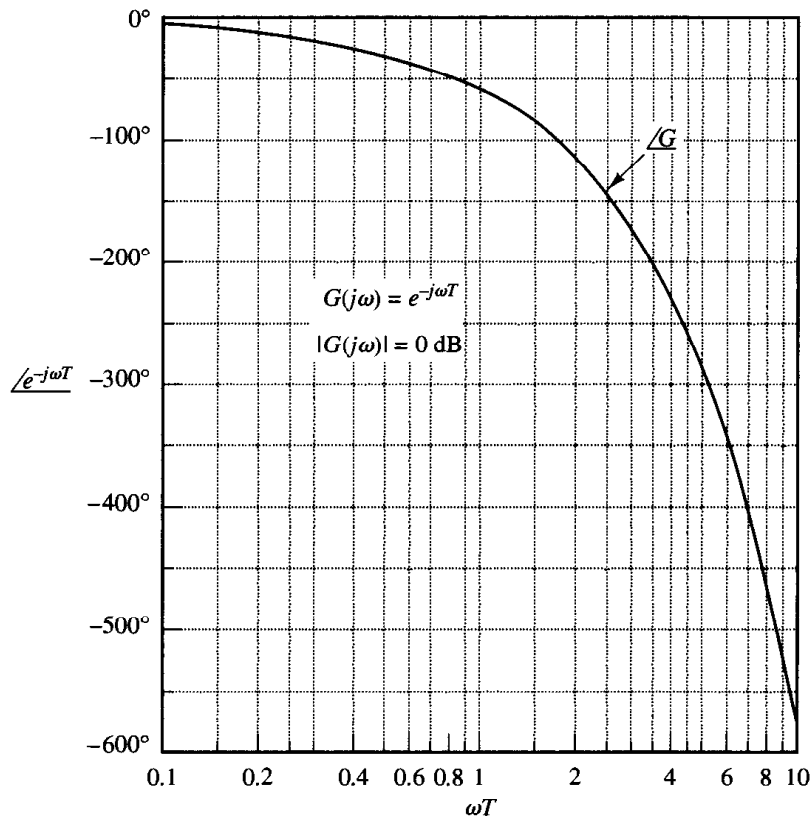
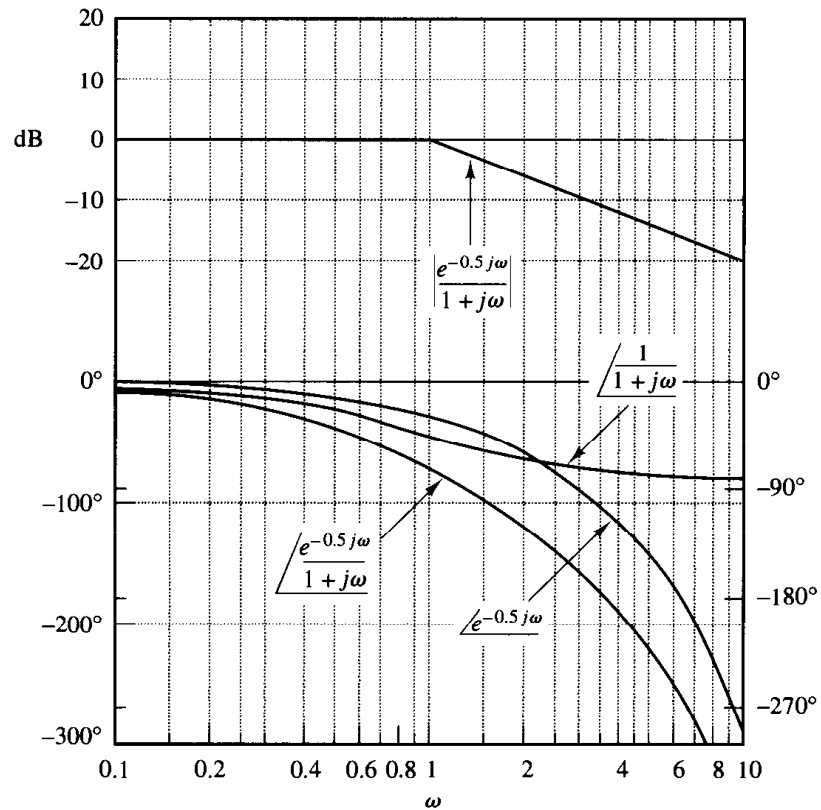


Figure 8–13
Phase-angle characteristic of transport lag.

Figure 8-14
Bode diagram for
the system $e^{-j\omega L}/$
 $(1 + j\omega T)$ with
 $L = 0.5$ and $T = 1$.



The log-magnitude and phase-angle curves for this transfer function with $L = 0.5$ and $T = 1$ are shown in Figure 8-14.

Relationship between system type and log-magnitude curve. Consider the unity-feedback control system. The static position, velocity, and acceleration error constants describe the low-frequency behavior of type 0, type 1, and type 2 systems, respectively. For a given system, only one of the static error constants is finite and significant. (The larger the value of the finite static error constant, the higher the loop gain is as ω approaches zero.)

The type of the system determines the slope of the log-magnitude curve at low frequencies. Thus, information concerning the existence and magnitude of the steady-state error of a control system to a given input can be determined from the observation of the low-frequency region of the log-magnitude curve.

Determination of static position error constants. Consider the unity-feedback control system shown in Figure 8-15. Assume that the open-loop transfer function is given by

$$G(s) = \frac{K(T_a s + 1)(T_b s + 1) \cdots (T_m s + 1)}{s^N (T_1 s + 1)(T_2 s + 1) \cdots (T_p s + 1)}$$

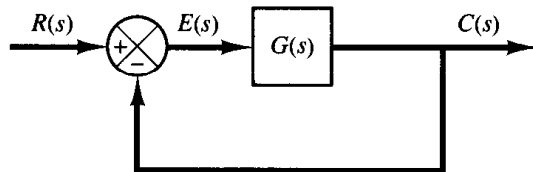


Figure 8-15
Unity-feedback control system.

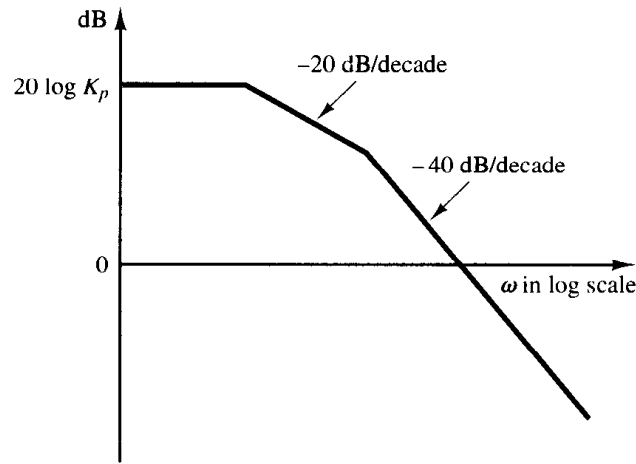


Figure 8-16
Log-magnitude curve of a type 0 system.

or

$$G(j\omega) = \frac{K(T_a j\omega + 1)(T_b j\omega + 1) \cdots (T_m j\omega + 1)}{(j\omega)^N (T_1 j\omega + 1)(T_2 j\omega + 1) \cdots (T_p j\omega + 1)}$$

Figure 8-16 shows an example of the log-magnitude plot of a type 0 system. In such a system, the magnitude of $G(j\omega)$ equals K_p at low frequencies, or

$$\lim_{\omega \rightarrow 0} G(j\omega) = K_p$$

It follows that the low-frequency asymptote is a horizontal line at $20 \log K_p$ dB.

Determination of static velocity error constants. Consider the unity-feedback control system shown in Figure 8-15. Figure 8-17 shows an example of the log-magnitude plot of a type 1 system. The intersection of the initial -20 dB/decade segment (or its extension) with the line $\omega = 1$ has the magnitude $20 \log K_v$. This may be seen as follows: In a type 1 system

$$G(j\omega) = \frac{K_v}{j\omega}, \quad \text{for } \omega \ll 1$$

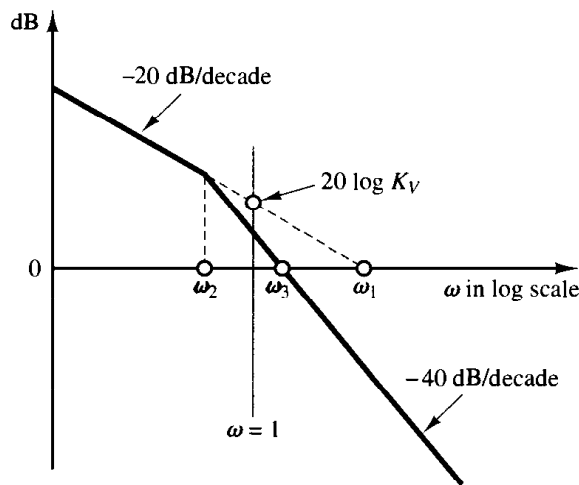


Figure 8-17
Log-magnitude curve of a type 1 system.

Thus,

$$20 \log \left| \frac{K_v}{j\omega} \right|_{\omega=1} = 20 \log K_v$$

The intersection of the initial -20 -dB/decade segment (or its extension) with the 0 -dB line has a frequency numerically equal to K_v . To see this, define the frequency at this intersection to be ω_1 ; then

$$\left| \frac{K_v}{j\omega_1} \right| = 1$$

or

$$K_v = \omega_1$$

As an example, consider the type 1 system with unity feedback whose open-loop transfer function is

$$G(s) = \frac{K}{s(Js + F)}$$

If we define the corner frequency to be ω_2 and the frequency at the intersection of the -40 -dB/decade segment (or its extension) with 0 -dB line to be ω_3 , then

$$\omega_2 = \frac{F}{J}, \quad \omega_3^2 = \frac{K}{J}$$

Since

$$\omega_1 = K_v = \frac{K}{F}$$

it follows that

$$\omega_1 \omega_2 = \omega_3^2$$

or

$$\frac{\omega_1}{\omega_3} = \frac{\omega_3}{\omega_2}$$

On the Bode diagram,

$$\log \omega_1 - \log \omega_3 = \log \omega_3 - \log \omega_2$$

Thus, the ω_3 point is just midway between the ω_2 and ω_1 points. The damping ratio ζ of the system is then

$$\zeta = \frac{F}{2\sqrt{KJ}} = \frac{\omega_2}{2\omega_3}$$

Determination of static acceleration error constants. Consider the unity-feedback control system shown in Figure 8–15. Figure 8–18 shows an example of the log-magnitude plot of a type 2 system. The intersection of the initial -40 -dB/decade

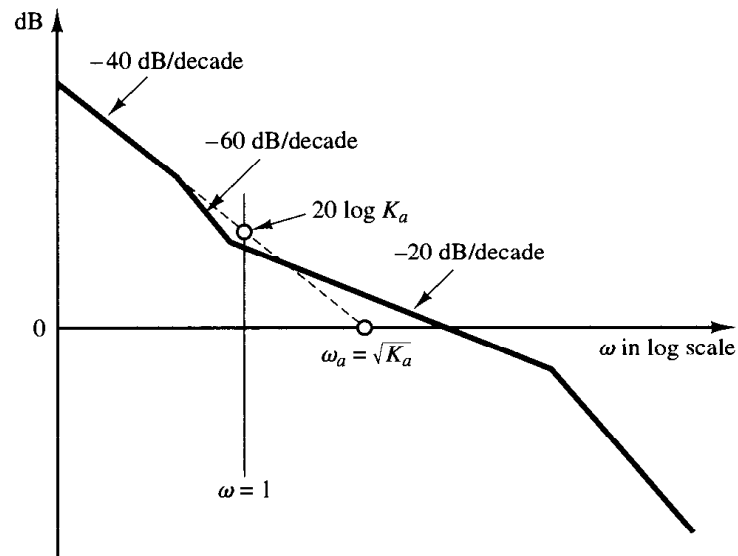


Figure 8-18
Log-magnitude curve
of a type 2 system.

segment (or its extension) with the $\omega = 1$ line has the magnitude of $20 \log K_a$. Since at low frequencies

$$G(j\omega) = \frac{K_a}{(j\omega)^2}, \quad \text{for } \omega \ll 1$$

it follows that

$$20 \log \left| \frac{K_a}{(j\omega)^2} \right|_{\omega=1} = 20 \log K_a$$

The frequency ω_a at the intersection of the initial -40 -dB/decade segment (or its extension) with the 0 -dB line gives the square root of K_a numerically. This can be seen from the following:

$$20 \log \left| \frac{K_a}{(j\omega_a)^2} \right| = 20 \log 1 = 0$$

which yields

$$\omega_a = \sqrt{K_a}$$

8-3 PLOTTING BODE DIAGRAMS WITH MATLAB

The command `bode` computes magnitudes and phase angles of the frequency response of continuous-time, linear, time-invariant systems.

When the command `bode` (without left-hand arguments) is entered in the computer, MATLAB produces a Bode plot on the screen.

When invoked with left-hand arguments,

$$[\text{mag}, \text{phase}, \text{w}] = \text{bode}(\text{num}, \text{den}, \text{w})$$

`bode` returns the frequency response of the system in matrices `mag`, `phase` and `w`. No plot is drawn on the screen. The matrices `mag` and `phase` contain magnitudes and phase

angles of the frequency response of the system evaluated at user-specified frequency points. The phase angle is returned in degrees. The magnitude can be converted to decibels with the statement

$$\text{magdB} = 20 \cdot \log_{10}(\text{mag})$$

To specify the frequency range, use the command `logspace(d1,d2)` or `logspace(d1,d2,n)`. `logspace(d1,d2)` generates a vector of 50 points logarithmically equally spaced between decades 10^{d1} and 10^{d2} . That is, to generate 50 points between 0.1 rad/sec and 100 rad/sec, enter the command

$$w = \text{logspace}(-1,2)$$

`logspace(d1,d2,n)` generates n points logarithmically equally spaced between decades 10^{d1} and 10^{d2} . For example, to generate 100 points between 1 rad/sec and 1000 rad/sec, enter the following command:

$$w = \text{logspace}(0,3,100)$$

To incorporate these frequency points when plotting Bode diagrams, use the command `bode(num,den,w)` or `bode(A,B,C,D,iu,w)`. These commands use the user-specified frequency vector w .

EXAMPLE 8-3

Consider the following transfer function:

$$G(s) = \frac{25}{s^2 + 4s + 25}$$

Plot a Bode diagram for this transfer function.

When the system is defined in the form

$$G(s) = \frac{\text{num}(s)}{\text{den}(s)}$$

use the command `bode(num,den)` to draw the Bode diagram. [When the numerator and denominator contain the polynomial coefficients in descending powers of s , `bode(num,den)` draws the Bode diagram.] MATLAB Program 8-1 shows a program to plot the Bode diagram for this system. The resulting Bode diagram is shown in Figure 8-19.

MATLAB Program 8-1

```
num = [0  0  25];
den = [1  4  25];
bode(num,den)
subplot(2,1,1);
title('Bode Diagram of G(s) = 25/(s^2 + 4s + 25)')
```

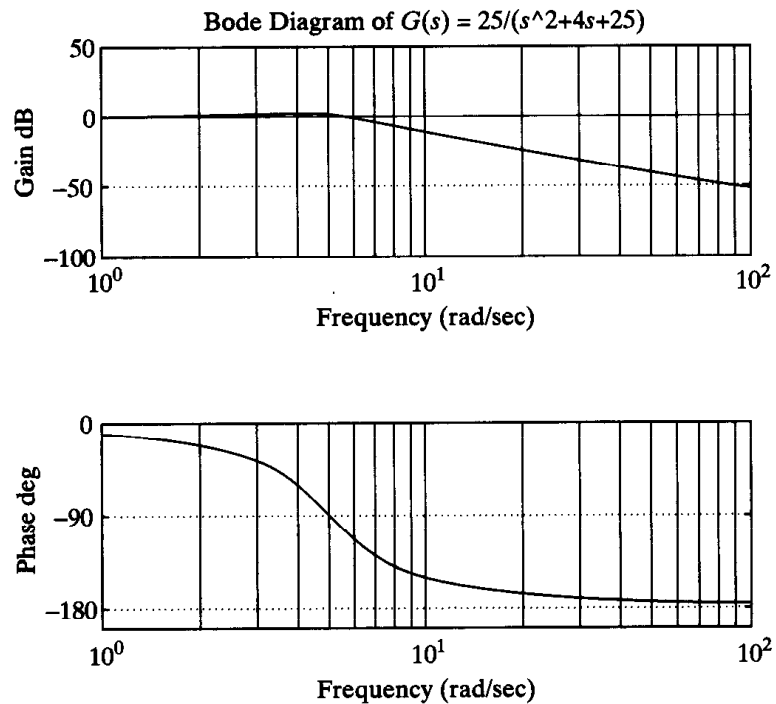


Figure 8–19
Bode diagram of
 $G(s) = \frac{25}{s^2 + 4s + 25}$.

EXAMPLE 8–4 Consider the system shown in Figure 8–20. The open-loop transfer function is

$$G(s) = \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}$$

Plot a bode diagram.

MATLAB Program 8–2 plots a Bode diagram for the system. The resulting plot is shown in Figure 8–21. The frequency range in this case is automatically determined to be from 0.1 to 10 rad/sec.

MATLAB Program 8–2

```
num = [0 9 1.8 9];
den = [1 1.2 9 0];
bode(num,den)
subplot(2,1,1);
title('Bode Diagram of G(s) = 9(s^2 + 0.2s + 1)/[s(s^2 + 1.2s + 9)]')
```

If it is desired to plot the Bode diagram from 0.01 to 1000 rad/sec, enter the following command:

$$w = \text{logspace}(-2,3,100)$$

Figure 8–20
Control system.

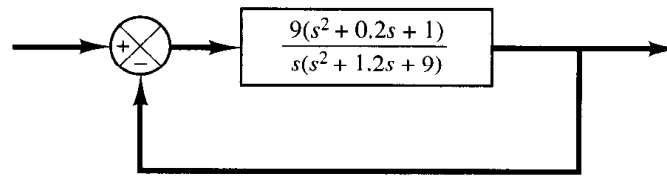
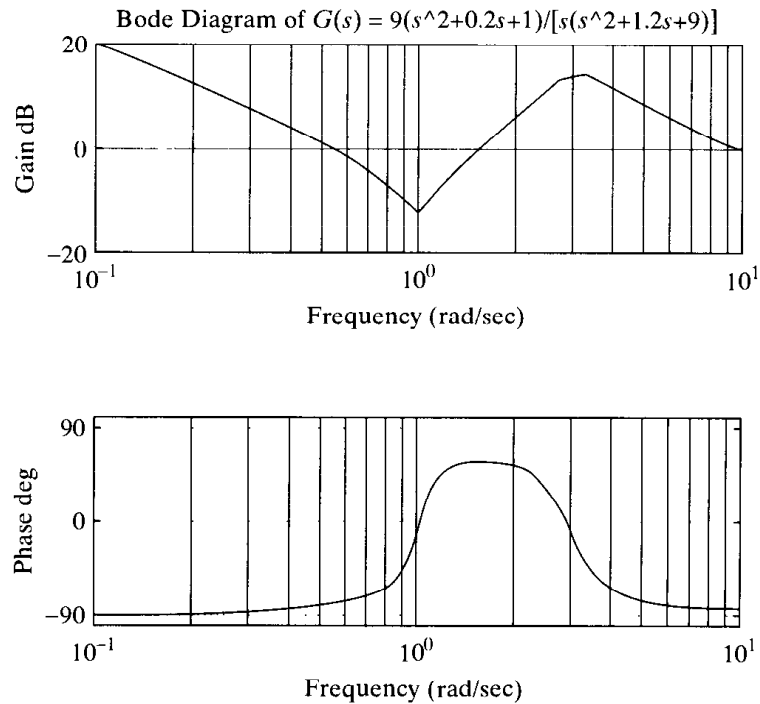


Figure 8–21
Bode diagram
of $G(s)$

$$= \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}$$



This command generates 100 points logarithmically equally spaced between 0.01 and 1000 rad/sec. (Note that such a vector w specifies the frequencies in radians per second at which the frequency response will be calculated.)

If we use the command

```
bode(num,den,w)
```

then the frequency range is as user specified, but the magnitude range and phase-angle range will be automatically determined. See MATLAB Program 8–3 and the resulting plot in Figure 8–22.

To specify the magnitude range and phase-angle range, use the following command:

```
[mag,phase,w] = bode(num,den,w)
```

The matrices mag and $phase$ contain the magnitudes and phase angles of the frequency response evaluated at the user-specified frequency points. The phase angle is returned in degrees. The magnitude can be converted to decibels with the statement

MATLAB Program 8-3

```
num = [0 9 1.8 9];
den = [1 1.2 9 0];
w = logspace(-2,3,100);
bode(num,den,w)
subplot(2,1,1);
title('Bode Diagram of G(s) = 9(s^2 + 0.2s + 1)/[s(s^2 + 1.2s + 9)]')
```

$\text{magdb} = 20 \cdot \log_{10}(\text{mag})$

If we wish to specify the magnitude range to be, for example, at least between -45 dB and $+45$ dB, then enter invisible lines at -45 dB and $+45$ dB in the plot by specifying dBmax (maximum magnitude) and dBmin (minimum magnitude) as follows:

```
dBmax = 45*ones(1,100);
dBmin = -45*ones(1,100);
```

Then enter the following semilog plot command:

```
semilogx(w,magdB,'o',w,magdB,'-',w,dBmax,'--i',w,dBmin,':i')
```

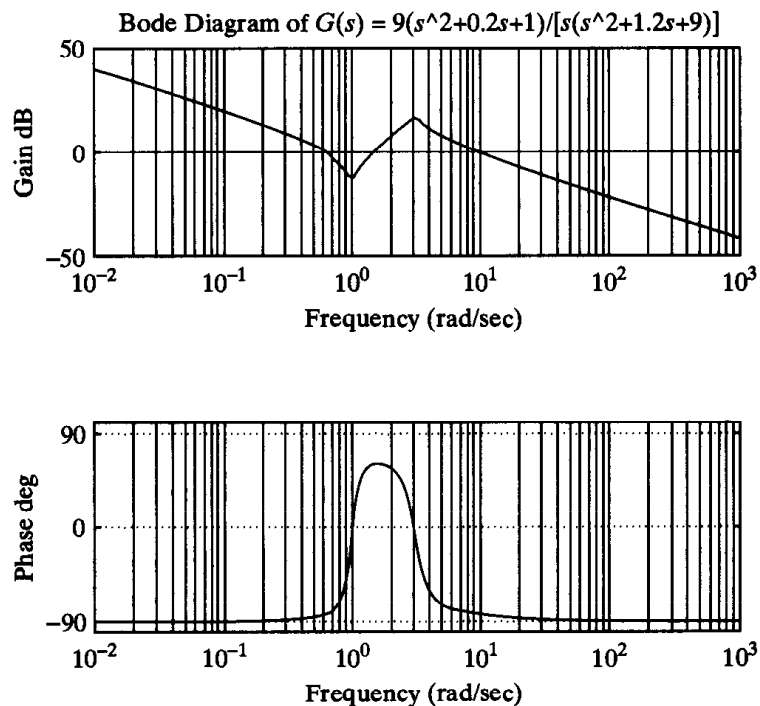


Figure 8-22

Bode diagram
of $G(s)$

$$= \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}$$

(Note that the number of dBmax points and that of dBmin points must be equal to the number of frequency points in w. In this example, all numbers are 100.) Then the screen will show the magnitude curve magdB with 'o' marks. (Straight lines at +45 dB and -45 dB are invisible.)

Note that 'i' is an invisible color. For example, 'og' will show small circles in green color and 'oi' will show small circles in 'invisible' color: that is, you will not see small circles in the screen. By changing a portion of the preceding semilogx command from

```
w,dBmax,'--i',w,dBmin,':i'
```

to

```
w,dBmax,'--',w,dBmin,':'
```

the +45-dB line and -45-dB line will become visible on the screen.

The range for the magnitude is normally a multiple of 5, 10, 20, or 50 dB. (There are exceptions.) For the present case, the range for the magnitude will be from -50 dB to +50 dB.

For the phase angle, if we wish to specify the range to be, for example, at least between -145° and +115°, we enter invisible lines at -145° and +115° in the program by specifying pmax (maximum phase angle) and pmin (minimum phase angle) as follows:

```
pmax = 115*ones(1,100)
pmin = -145*ones(1,100)
```

Then enter the semilog plot command:

```
semilogx(w,phase,'o',w,phase,'-',w,pmax,'--i',w,pmin,':i')
```

(The number of pmax points and that of pmin points must be equal to the number of frequency points in w.) The screen will show the phase curve. Straight lines at +115° and -145° are invisible.

The range for the phase angle is normally a multiple of 5°, 10°, 50°, or 100°. (There are exceptions.) For the present case, the range for the phase angle will be from -150° to +150°.

MATLAB Program 8-4 produces the Bode diagram for the system such that the frequency range is from 0.01 to 1000 rad/sec, the magnitude range is from -50 to +50 dB (the magnitude range is a multiple of 50 dB), and the phase-angle range is from -150° to +150° (the phase-angle range is a multiple of 50°). Figure 8-23 shows the Bode diagram obtained by use of MATLAB Program 8-4.

What happens to the Bode diagram if the gain becomes infinite at a certain frequency point? If there is a system pole on the $j\omega$ axis and the w vector happens to contain this frequency point, the gain becomes infinite at this frequency. In such a case, MATLAB produces warning messages. Consider the following example.

MATLAB Program 8–4

```
% ----- Bode diagram -----

% ***** In this program we shall obtain Bode diagram of
% transfer-function system using user-specified frequency
% range *****

% ***** Enter the numerator and denominator of the transfer
% function *****

num = [0 9 1.8 9];
den = [1 1.2 9 0];

% ***** Specify the frequency range and enter the command
% [mag,phase,w] = bode(num,den,w) *****

w = logspace(-2,3,100);
[mag,phase,w] = bode(num,den,w);

% ***** Convert mag to decibels *****

magdB = 20*log10(mag);

% ***** Specify the range for magnitude. For the system
% considered, the magnitude range should include -45 dB
% and +45 dB. Enter dBmax and dBmin in the program and
% draw dBmax line and dBmin line in invisible color. To
% plot the magdB curve and invisible lines, enter the
% following dBmax, dBmin, and semilogx command *****

dBmax = 45*ones(1,100);
dBmin = -45*ones(1,100);
semilogx(w,magdB,'o',w,magdB,'-',w,dBmax,'--i',w,dBmin,':i')

% ***** Enter grid, title, xlabel, and ylabel *****

grid
title('Bode Diagram of  $G(s) = 9(s^2 + 0.2s + 1)/(s^2 + 1.2s + 9)$ ')
xlabel('Frequency (rad/sec)')
ylabel('Gain dB')

% ***** Next, we shall plot the phase-angle curve *****

% ***** Specify the range for phase angle. For the system
% considered, the phase-angle range should include -145 degrees
% and +115 degrees. Enter pmax and pmin in the program and
% draw pmax line and pmin line in invisible color. To plot
% the phase curve and invisible lines, enter the following
% pmax, pmin, and semilogx command *****
```